

The Information Content of a Limit Order Book: the Case of an FX Market

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Abstract

In this paper we examine the question of whether knowledge of the information contained in a limit order book helps to provide economic value in a simple trading scheme. Given the greater information content of the order book, over simple price information, it might naturally be expected that the order book would dominate. The literature however is divided and it is still an open question as to whether the order book adds value. We find that despite the statistical significance of variables describing the structure of the limit order book in explaining tick by tick returns, they do not consistently add significant economic value. We show this using a simple linear model to determine trading activity as well as a model free genetic algorithm based on price, order flow and order book information. Using Dollar Sterling tick data we find that the greater transparency provided by knowledge of the structure of the limit order book does not translate into greater profitability beyond that provided by price information alone. We also find that the profitability of all trading rules based on genetic algorithms dropped substantially in 2008 compared to the 2003 data.

Key words: profitability, limit order book, high-frequency data.

JEL Classification: D81, F31, C53

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1 Introduction

One important issue in recent market microstructure research has been whether knowledge of the structure of the limit order book is informative regarding future price movements. There is a growing body of theoretical work suggesting that limit orders imply the predictability of short term asset returns (see Handa and Schwartz (1996), Harris (1998), Parlour (1998), Foucault (1999), Rosu (2010) among others). This is in contrast with earlier papers that implied that informed traders would only use market orders (see Glosten (1994), Rock (1996) and Seppi (1997)). This debate has also been carried out empirically (Harris and Hasbrouck (1996), Kavajecz (1999), Harris and Panchapagesan (2005), Cao, Hansch and Wang (2009), Rime, Sarno and Sojli (2010)) all of whom demonstrated that asset returns can be explained by limit order book information, such as depth and order flow. However, these studies have failed to demonstrate that the predictability of returns can be exploited in economic terms. In this paper we go beyond statistical significance and consider the economic value of limit order book information in the FX market.

We address this question by explicitly constructing trading strategies based on full limit order book and price information in the FX market. These strategies only use historical information in order to ensure that trading can be implemented in “real time” and focus on the economic value of ex-ante predictability in out-of-sample prediction exercises.

Such an approach requires the explicit specification of a forecasting model which might be misspecified and so spuriously unable to exploit the available information. In order to consider these issues we construct trading strategies in a model-free way by employing a genetic algorithm. Genetic algorithms serve as a search mechanism for the best trading rule from amongst a huge universe of potential rules given the particular information set and have been successfully applied in a number of financial applications, most notably Dempster and Jones (2001), Dworman, Kimbrough and Laing (1996), Chen and Yeh (1997b), Chen and Yeh (1997a), Neely, Weller and Dittmar (1997), Allen and Karjalainen (1999), Neely and Weller (2001), Chen, Duffy and Yeh (1999), Arifovic (1996). Rather than relying on a specific forecasting model, the genetic algorithm searches for a trading rule which exploits the information most efficiently. We then test if this trading rule generates significantly higher returns when new information constructed from the limit order book is included alongside price information.

It is important to recognize the theoretical and practical coherence offered by using the genetic algorithm approach. A number of authors, since Leitch and Tanner (1991), have argued that the use of purely statistical criteria to evaluate forecasts and

trading strategies is inappropriate (see, for instance, Satchell and Timmermann (1995), Granger and Pesaran (2000), Pesaran and Skouras (2002) and Granger and Machina (2006)). The issue turns on the appropriate loss function and whereas many statistical evaluation criteria are based on a quadratic loss, practical criteria are more likely to be based on the utility derived from profits. Critically from our point of view the genetic algorithm constructs trading rules using the same loss function as is used to evaluate the out-of-sample performance of the trading strategy unlike the linear regression model where a statistical quadratic loss is used in estimation.

Another important factor that needs to be considered when testing the profitability of trading strategies in “real time” is transaction costs.¹ We analyze the performance of our trading rules on the basis of the best bid and ask prices using tick-by-tick data and so explicitly take into account transaction costs as measured by the bid-ask spread. This allows us to test if predictable components in exchange rate returns are economically exploitable net of transaction costs.

Using data on the GBP/USD exchange rate for five separate weeks, we find statistical predictability in the exchange rate and profitability net of transaction costs for samples drawn from 2003. However, we find that the profitability in more recent data from 2008 decreases substantially and in most cases is not significantly different from zero. This could be explained by the tremendous recent growth in high-frequency algorithmic trading within financial markets.

We also find in-sample statistical significance of limit order book information in all sample periods. Specifically, we show that both static information about liquidity beyond the best prices and order flow of both market and limit orders have some ability to explain future short-term movements of the exchange rate. However, we find little or no value in an economic sense in allowing the predictor to exploit information in the order book beyond of that contained in the best prices. We fail to significantly increase out-of-sample returns from our trading strategy when we use liquidity and order flow information. Our main finding then is that any information contained in limit orders beyond best prices is not robust enough to be exploited profitably out-of-sample, particularly in the most recent data.

The remainder of the paper is organized as follows. In Section 2 we provide a literature review relevant to this research. Section 3 contains a description of the data used in the study and the methodology employed in the analysis. The main results are given in Section 4. Section 5 concludes.

¹Neely and Weller (2003) for instance emphasise the critical role of transaction costs and inconsistencies between the data used by practitioners and in academic simulations.

2 Literature Review

Limit-order book markets potentially offer greater transparency when compared with quote-driven markets. Whereas dealer markets will usually only release the dealers' best quotes, a limit-order-book can allow its users to view the depth at a number of price levels away from the market price. The NYSE, under what is known as the OpenBook program began to publish aggregate depths at all price levels on either side of the book in 2002 and under LiquidityQuote will display a bid and offer quote, potentially different from the best quotes in the market. NASDAQ's SuperMontage displays aggregate depths at five best price levels on either side and employs a scan function which allows traders to assess liquidity further along the book. The question is how this incremental information on the structure of a limit order book is used and whether it adds economic value in the process of price discovery.

There are two closely related literatures which bear on the research we report below. The first considers the choice of order type, market or limit order, and how the structure of the limit order book and traders' preferences affect this choice. The second considers whether the structure of the limit order book is informative regarding the evolution of future prices.

Theoretical papers by Glosten (1994), Rock (1996) and Seppi (1997) assume that informed traders use market orders so limit orders beyond the best prices can contain little information. Informed traders enter the market aggressively to exploit their private information using market orders with guaranteed immediate execution. Chakravarty and Holden (1995) consider a model where informed traders are allowed to submit both limit and market orders and show that optimal order placement strategy consists of a combination of limit and market orders. Parlour (1998) presents a dynamic model of a limit order market in which the decision to submit a market order or a limit order depends on the current state of liquidity and the trader's place in the limit order queue. Foucault (1999) presents a dynamic model of limit/market order placement in a market with heterogenous asset valuation among traders with no private information. He showed that there is a trade-off between limit and market order strategies which depends on the volatility of asset returns and bid-ask spreads. Handa and Schwartz (1996) and Handa and Schwartz (2003) examine the impact of asymmetric information on the order placement strategies. They show that if the cost of being picked-off by an informed trader is lower than the expected gain to limit order execution then a limit order strategy can be profitable. Rinaldo (2004) examines how the state of the limit order book affects a trader's strategy. He shows that patient traders become more aggressive and hence use more market orders when their side of the book is thicker,

the spread wider, and the volatility increases. More recently, Kaniel and Liu (2006) present a simple equilibrium model to investigate whether informed traders use limit or market orders. They show that informed traders prefer limit orders when the private information is long-lived, and hence limit orders convey more information than market orders. When this is the case and the number of traders who can discover the private information is small then using market orders will reveal too much information implying higher trading costs.

A number of papers have considered the empirical determinants of order aggressiveness. In particular, Bloomfield, O'Hara and Saar (2005), in a laboratory experiment, find that informed traders submit more limit orders than market orders. They exploit their informational advantage early in the trading period to find mispriced limit orders moving the market towards the true price, and thereby progressively reduce the value of their information. As the end of the trading period approaches, they switch increasingly to limit orders, as the value of their informational advantage falls away. Cao et al. (2009) considered the information content of a limit-order book behind the best bid and offer using data from the Australian Stock Exchange. They found that the contribution of the order book to price discovery is approximately 22% while the rest of the variation in future returns comes from the best bid, offer and transaction prices. They demonstrate that order imbalances between the demand and supply schedules along the book are statistically significantly related to future short-term returns. Harris and Panchapagesan (2005) also found that information on limit order book depth forecasts short term changes in prices.

Hellstroem and Simonsen (2009) using a count data time series approach find that there is informational value in the first levels of the bid- and ask-side of the order book. They also show that both the change and the imbalance of the order book statistically significantly explain future price changes. Offered quantities at the best bid and ask prices on data from the Swedish Stock Exchange reveal more information about future short run returns than measures capturing the quantities at prices below and above. The impacts are most apparent at 1 minute aggregation level while results for higher aggregation levels generally showed insignificant results. These results would suggest that the informational content of the order book is very short-term.

There are also several papers demonstrating some ability of order-flow to predict future movements of exchange rates. Evans and Lyons (2005) and Evans and Lyons (2006) were among first to document the forecasting power of customer order flow to outperform a random walk benchmark. Froot and Ramadorai (2005) report that order flow contains some information about future exchange rate returns in low frequency data. Rime et al. (2010) employ data for three major exchange rates from the Reuters

electronic interdealer trading platform and confirm these findings. In contrast to the above studies, Danielsson, Luo and Payne (2002) find limited and Sager and Taylor (2008) find no evidence of superior forecasting ability of order flow over random walk models at different forecast horizons. However, a very recent paper by Latza and Payne (2010) has considered the forecasting power of market and limit order flows on stock returns and show that both can forecast returns. They also find the predictive power of limit order flows was greater than that of market order flows and greater for more liquid stocks.

This brief review indicates that there are a range of results in both the theoretical and empirical literatures as to whether limit orders convey information beyond the best bid and ask prices. However, there are few studies that focus whether this information can be efficiently and systematically exploited by market participants (with two exceptions, Della Corte, Sarno and Tsiakas (2009) and Rime et al. (2010), who only study profitability of transaction order flow in the long run).

Where empirical results have shown statistical significance on future returns (which may be transitory) the economic value of incorporating information on the structure of the limit order book does not seem to have led to greater economic profits through trading rules that exploit this information. Although Latza and Payne (2010) do offer some recent evidence in this respect. It is this question that we now turn to address.

3 Data and Methodology

We use interdealer tick by tick data for the US dollar pound sterling exchange rate drawn from the Reuters D3000 trading system, which is the platform where most GBP trades take place. In order to make sure that our results are not driven by the use of any particular sample period, we use five different data sets: weeks commencing on 13 January 2003, 10 February 2003, 17 March 2003 and two days on 31 March and 1 April 2008. The data we analyze consists of continuously recorded limit and market orders and their volumes between 07:00-17:00 GMT which allows us to reconstruct the full limit order book on a tick-by-tick basis. For each entry, the data set contains a unique order identifier, quoted price, order quantity, quantity traded, order type, transaction identifier of order entered or removed, status of market order, entry type of orders, removal reason, date and time of orders entered and removed. The data time stamp's precision is $1/100^{\text{th}}$ of a second and the minimum trade size in Reuters electronic trading system is 1 million pounds sterling.

3.1 Summary Statistics

Table 1 reports the summary statistics for transaction and limit order data for the different sample periods. Average inter-quote durations (speed of limit order arrival or removal) are 2.29, 1.82 and 1.84 seconds for first three samples and 0.1 and 0.09 seconds for the two 2008 samples. This demonstrates that the electronic market is very active and critically its activity has grown tremendously from 2003 to 2008. There are 75,135, 98,785 and 97,559 orders for the three weeks in 2003 and 594,519 and 388,259 on 31 March and 1 April 2008 respectively. The average values of the bid-ask spread from our samples are 2.28, 2.21, 2.73 basis points in 2003 and 2.53, 2.59 basis points in 2008, indicating that D3000 is a very tight market.

Although there is no major difference in the bid-ask spreads between the 2003 and 2008 samples, there is a huge jump in market liquidity as measured by the slope of the limit order book and its depth. The average slopes of the limit order book in the 2003 subsamples are 58.69, 55.53 and 75.54 basis points per billion of currency trade for the bid side and 65.35, 66.71 and 85.03 for the ask side.² In the 2008 samples, the slope values were 20.89 and 19.52 for bid side and 21.52 and 20.26 for ask side, indicating that the limit order book became about 3 times flatter in 2008 than it was in 2003. Also, the depth of the market almost doubled. These summary statistics indicate that the currency pair we are studying is traded in a highly liquid market.

Insert Table 1 here

3.2 Testing Hypotheses

We are interested in two main questions. The first is whether the exchange rate is predictable at the highest possible frequency and if this predictability translates into statistically significant economic profits. Thus,

Hypothesis 1. The exchange rate is not predictable at high frequency.

The second question is whether limit order book information adds any economic value to the predictability of the exchange rate beyond that provided by publicly-available price information. Hence,

Hypothesis 2. Limit order book information does not add value to the predictability of the exchange rate at high-frequency.

As mentioned above, we build trading strategies which are designed to exploit any profitable pattern in the exchange rate and test the profits obtained for significance.

²We construct the slope of the demand and supply curves in our limit order book using the two best bid and ask quotes and the associated depth at these quotes. See also section 3.3 for detailed description of variables.

By varying the information set used in these trading rules we are able to differentiate the predictive power of limit order book information from that contained in past prices and volumes.

As the majority of existing research has focused on linear predictive models, we also employ a *linear model* to forecast future exchange rate movements as a benchmark. Apart from the linear model we use a *genetic algorithm* as a general non-parametric device to construct trading rules. This approach has the advantage as it is *model free* and is designed to exploit both linear and any *non-linear* dependency between future returns and predictors. The specification of the genetic algorithm trading rules evolve according to their “fitness ” which is determined by an economic profit based criterion. This approach is not as susceptible to the criticism that any result we found could have been due to the structure of a specific trading rule we had selected ex ante. A disadvantage of the genetic algorithm based strategies is that they lose the simple interpretation that a linear model retains.

Since we want to keep our trading strategies implementable in “real time” we need to ensure that they are based exclusively on historical and available information. We use an in-sample period to construct the rules and check their performance out-of-sample and all conditioning information is based on past variables prior to the time of decision making in the chosen trading rule.

We describe the implementation of these two approaches more precisely in the following sections.

3.3 The Information Sets

We define four different conditioning information sets in order to test for the added value of various limit order book variables. Each of these information sets contain variables which are available at each point of time to the trader. The values of these variables serve as inputs to a function (either the linear rule or the genetic algorithm) which generates the trading signal.

1. *Screen information* (denoted by *Screen* hereafter) contains best limit order prices (both bid and ask) and their quantities as time series, the bid-ask spread, the level of mid-quotes and the inter-quote duration. This information is considered as the basic set which is normally available to all traders. It is also contained as a subset in the three further information sets defined below.
2. *Limit order book information* (denoted by *Book* hereafter) is an extension of the *Screen* information. In addition to the variables mentioned above, it includes total depth, the number of layers in the limit order book, the difference between

the best and the second best price (both, bid and ask), the slopes of the bid and ask curves of the limit order book, time series of the levels of quantity weighted quotes and the quantity weighted mid-quote, the quantity weighed bid-ask spread and the difference between the mid-quote and the quantity weighted mid-quote. By depth we mean the total quantity available at the moment in the limit order book on the particular side of the book (demand or supply). The slope of the bid side of the limit order book is defined as

$$slope^{bid} = \frac{p_1^{bid} - p_2^{bid}}{q_1^{bid}},$$

where p_1^{bid} and p_2^{bid} are the best and the second best prices on the bid side respectively and q_1^{bid} is the quantity available at the best bid price. The slope of the ask schedule is defined analogously. The quantity weighted bid price is defined as

$$wp^{bid} = \frac{\sum_i (p_i^{bid} \times q_i^{bid})}{\sum_i q_i^{bid}},$$

where the index i runs through all available levels of bid quotes. The quantity weighted mid-quote is

$$wmid = \frac{wp^{bid} + wp^{ask}}{2}$$

and the quantity weighted bid-ask spread is

$$wspread = wp^{ask} - wp^{bid}.$$

3. *Order flow information* (denoted by *Order* hereafter) contains the screen information set plus order flow information. Following Latza and Payne (2010), we use two different types of order flow: limit order flow and transaction order flow. Limit order flow is further decomposed into order flow on the best prices (the inside order flow) and order flow outside the best prices (the outside order flow). We construct 1 and 20 minute as well as 1 tick order flow variables for each side of the limit order book. By 1 tick order flow we mean the volume of the most recent order of the corresponding type (either market order, limit order at the best price or the limit order outside the best price) strictly preceding the time of decision making.
4. *Full information* (denoted by *Full* hereafter) combines all three types of conditioning information mentioned above.

The Reuters D3000 trading platform did not allow traders to see limit orders and their quantities outside the best bid and ask prices in 2003 and 2008. Hence it would be impossible for traders to build strategies that explicitly use this information. Thus, if the three extended information sets are found to convey substantial information which can be profitably exploited, we will detect it in our experiments. If we cannot find added value, then it would imply that limit orders placed outside the best prices do not carry any significant information about future returns.

3.4 The Trading Mechanism and Fitness Function

We measure the fitness of trading rules by means of the cumulative returns from the following simple trading strategy. The trader buys or sells short 1 million pounds sterling according to the signal provided by the selected trading rule. This allows us to control for the potential price impact of trade since we can ensure that liquidity necessary to complete transaction with the minimum trade size is present in the market.³ As new information arrives from the market, the trader re-evaluates the trading signal and updates his position accordingly. This means that as soon as the trader observes any change in the limit order book, he can change or keep the same position depending on the outcome of the signal.

Under such a trading scheme, the trader is potentially able to trade at every single instant. In order to control for trading frequency we add a trading threshold to the strategy. According to this, the trader is allowed to trade only if the exchange rate level exceeds a band of $\pm k$, relative to his last transaction price. More formally, let z_t denote the state of the investor's position at time t . That is, $z_t = 1$ corresponds to a long position in sterling and $z_t = -1$ corresponds to a short position. The trader will re-evaluate his position only if $|e_t - e_{t_1}| \geq k$, where t_1 denotes time of the trader's last transaction.

The coefficient k serves as an inertia parameter to filter out weak trading signals. The idea of such "filter rules" goes back to Alexander (1961) and Fama and Blume (1966). The parameter k determines an "inertia band" that prompts one to trade only once a realization of the exchange rate exceeds the value of a certain characteristic (past realized values of the exchange rate in our case) by a value of k . A larger inertia band (larger k) filters out more trades, thus reducing trading frequency. One interpretation of this market timing rule is based on a belief that there is no particular directional abnormality if the value of the particular variable is not statistically significant. The use of an inertia parameter also has a behavioural interpretation based on the notion

³We assume that traders can execute transactions at the current price immediately and are not affected by either latency problems nor execution risk.

of ambiguity aversion. Easley and O'Hara (2010) show that in the face of Knightian uncertainty incomplete preferences may lead to an absence of trading. Traders will revise their position only if the trading signal is confirmed by other criteria that they have at their disposal, which is very often provided by simple technical tools.

Table 2 presents how many times the trader re-evaluates the position during the out-of-sample period for different values of k . A zero value of k means that trades can take place every time the mid-quote of the exchange rate changes and hence exhibits the largest number of transactions. As k increases, the trading frequency drops. For $k = 30$ only up to 10 transactions per day can be made.

Insert Table 2 here

We use simple cumulative returns as a performance measure to evaluate the profitability of trading strategies:

$$R_c = \prod_t (1 + z_t r_t) - 1,$$

where $r_t = \frac{e_t - e_{t-1}}{e_{t-1}}$ is the one-period return of the exchange rate e_t . Here e_t is the mid-quote exchange rate when transaction costs are not taken into account; otherwise e_t denotes the corresponding best bid or ask price.

3.5 The Linear Trading Rule

The linear model's predictions are generated using a linear regression. We use an in-sample period to estimate the regression model where the dependent variable is the one step ahead mid-quote exchange rate return r_{t+1} and the regressors are time t dated values of all the variables contained in the relevant information set. Out-of-sample forecasts of future exchange rate returns serve as signals for a simple binary trading rule, i.e. positive (negative) predicted values of future returns are associated with a "buy" ("sell") signal. Based on these signals we construct a trading strategy as described above and evaluate its out-of-sample performance for different values of the inertia parameter k .

3.6 Genetic Algorithm Trading Rule

The genetic algorithm provides an effective method for searching over a huge space of potential trading rules, both linear and non-linear. This method allows us to evaluate predictability as generally as possible and not to impose any effective restriction on the form of the model, predictor or trading rule. The genetic algorithm is a computer

based optimization procedure which uses the evolutionary principle – the survival of the fittest – to find an optimum. It provides a systematic search process directed by performance rather than gradient.⁴

Starting from an initial set of rules, the genetic algorithm evaluates the fitness of various candidate solutions (trading rules) using the given objective function. It provides as an output, solutions that have higher in-sample cumulative returns on average.

We build a trading rule as a binary logical tree which produces true or false signals given the set of input variables. If the value of the rule is “true”, it gives the signal to “buy” an asset. If the rule is “false” – the trader “sells” the asset short. The rules are represented in the form of randomly created binary trees with terminals and operations in their nodes. We employ the following choices of operations and terminals.

Operations: the function set used to define the technical rules consists of the binary algebraic operations $\{+, -, *, /, \max, \min\}$, binary order relations $\{<, >, \leq, \geq, =\}$, logical operations $\{\text{and}, \text{or}\}$ and unary functions $\{\text{abs}, -\}$ of absolute value and change of sign.

Terminals: the terminal set contains the variables, which take their values from data and are updated every time new information arrives in the market. Thus, it allows the conditioning information sets to update the trading rule as time passes. The genetic algorithm also explicitly computes lag values of the conditioning variables, their moving average values and maxima and minima over different periods. The terminal set also includes real numbers as terminal constants.

An example of a tree and the corresponding trading strategy is given in Figure 1. It presents a trading rule which generates a signal “buy” if the current quantity weighted spread is less than the last ten trades average returns times the bid-ask spread and the absolute value of the difference between the quantity-weighted bid and best bid price is less than 0.001. Otherwise, the signal is “sell”.

Insert Figure 1 here

Two operations of crossover and mutation are applied to create a new generation of decision rules based on the genetic information of the fittest candidate solutions.

⁴Nix and Vose (1992) and Vose (1993) use a Markov Chain framework to show that asymptotically in population size – populations that have suboptimal average fitness have probabilities approaching zero in the stationary distribution of the Markov Chain whereas the probability for the population that has optimal average fitness approaches one. The genetic algorithm’s success as an optimizer depends on having a sufficiently large population of individual rules. We have taken considerable effort to ensure that this is the case in our experiments with the results repeated below representing only a fraction of a very large computational exercise.

Crossover For the crossover operation one randomly selects two parents from the population based on their fitness, then take a node within each parent as a crossover point selected randomly and the subtrees at the selected nodes are exchanged to generate two children. One of the offspring then replaces the less fit parent in the population. In our implementation we use a crossover rate of 0.4 for all individuals in the population. This operation combines the features of two parent chromosomes to form two similar offspring by swapping corresponding segments of the parents. In our case these segments are represented by sub-nodes of a binary tree. The intuition behind the crossover operator is information exchange between different potential solutions.

Mutation In order to mutate a rule one of its subtrees is selected at random and replaced with the new randomly generated tree. This operation guarantees the refreshment of the genetic code within the population. The best 25% of rules are not mutated at all and the remaining are mutated with probability 0.1. The intuition behind the mutation operator is the introduction of some extra variability into the population of trading rules.

The evolutionary algorithm can be summarized as follows:

1. Create randomly the initial population $P(0)$ of trading rules given the information set and initialize the number of iterations $i = 0$;
2. Set $i := i + 1$;
3. Evaluate in-sample fitness of each tree in the population using the fitness function;
4. Generate a new population of trees (i.e. the set of new trading rules) using the genetic operations (crossover and mutation) and replace the old population with the new one;
5. Repeat 2)–5) while $i < N$.

After each such iteration, rules which have poor performance according to the fitness function are removed from the population and only the more profitable candidates survive and carry their structure onwards to create new trading rules. Ultimately, the algorithm converges to the trading rule achieving the best in-sample performance given the conditioning information.

In the program we have experimented and use a population size of 200 individual trading rules and perform 1000 iterations of the algorithm (that is, $N = 1000$).

The complexity of trading rules is controlled in a probabilistic manner. In fact, the probability for a binary node to appear in the tree is smaller than the probability of a unary one which prevents the tree to becoming very large.

3.7 Testing Procedures

To test for profitability of each of the trading strategies we employ a series of statistical tests as follows. We split each trading period into two equal parts which serve as in-sample and out-of-sample periods respectively. We use the in-sample period as the estimation sample for the linear regression model. We test the economic value of a strategy using Anatolyev-Gerko statistic (see Anatolyev and Gerko (2005)). This test compares the profitability of a trading strategy relative to the random walk model. Relative performance of trading strategies based on different conditioning information sets are then tested using Giacomini-White test for conditional predictive ability (see Giacomini and White (2006)).

Similarly, with the genetic algorithm based strategy, we choose the trading rule which produces the best in-sample performance and test its profitability out-of-sample. In order to generate an empirical distribution of the out-of-sample cumulative returns we run this procedure independently 100 times. This provides us with potentially 100 different trading rules (due to the stochastic nature of the GA search) and their out-of-sample performance. Using this sample of independent cumulative out-of-sample returns we can use a t-statistic to test if the mean of the returns is significantly different from zero. The relative performance of the different information sets is tested using a paired t-test. Specifically we test if the difference in the unconditional mean of returns for two strategies based on different information sets is significantly different from zero.

In addition, we combine signals from the 100 best individual trading rules and create an aggregated genetic algorithm signal, which we call the “Majority” rule. This combined rule is an alternative strategy to the single best in-sample genetic algorithm rule. The “Majority” rule produces a “buy” (“sell”) signal if the majority of the 99 independent best in-sample rules produce the “buy” (“sell”) signal. This rule probably reflects the way in which technical analysis is used by practitioners. Traders often do not follow a single rule but form an impression as to where the market is moving on the basis of a number of technical indicators, dropping those that appear not to have worked well in the past. The economic value of this rule is then tested using the Anatolyev-Gerko test and the relative performance is tested by the Giacomini-White test.

We carry out two separate exercises by allowing the trader to trade on either mid-quotes (without transaction costs) or on bid and ask prices (taking into account transaction costs explicitly). In the former case, the trader always makes a transaction at the mid-quote regardless of whether he buys or sells. In the latter, the trader always buys at the best ask price and sells at the best bid price. Thus, the current bid-ask spread reflects the real transaction costs a trader would face in the market.

4 Results

We now turn to discuss the empirical results from our experiments. We start by examining Hypothesis 1 as to whether there is evidence for the predictability and profitability of exchange rates at high frequency and then we consider the relative performance of the different information sets.

4.1 Hypothesis 1: Predictability and Profitability

In-sample coefficient estimates for different information sets for the linear model are given in Table 3.

Insert Table 3 here

The results show that for each of the three extended information sets beyond *Screen* information the majority of the explanatory variables are statistically significant. Moreover, we can easily reject the joint hypothesis that all coefficients in each of the extended information sets beyond the *Screen* information set are insignificant from zero on the basis of an F-test. This confirms the results in the existing literature that statistically, limit order book information does contribute to the in-sample explanation of the exchange rate. In order to check whether this apparent predictability can be translated into out-of-sample profitability we implement the trading strategy described above using the in-sample coefficients estimates.

4.1.1 No Transaction Costs

Table 4 reports the returns⁵ of the linear strategy at different values of k under no transaction costs.

Insert Table 4 here

We first examine the linear trading rule using the Anatolyev-Gerko test for superior predictability relative to a random walk. For the 2003 samples, out-of-sample returns are positive and statistically superior to the random walk for inertia parameters k ranging from 0 to 6 basis points. Returns remain positive but are not systematically superior to the random walk for k values up to 16 basis points. Once $k \geq 18$ we can find systematically negative returns. Out-of-sample returns for the 2008 sample periods decrease substantially and remain positive across all information sets only for k values equal to 0 and 2 basis points. Moreover, the linear model outperforms the

⁵All returns presented in this paper is adjusted to a daily basis.

random walk only for $k = 0$ as indicated by Anatolyev-Gerko statistics. The highest returns are achieved at the lowest value of k across all samples simply because of the large number of transactions performed with no transaction costs.

Similar results are produced by the genetic algorithm strategy. Table 5 reports the average out-of-sample performance of the best 100 genetic algorithm trading rules for the five sample periods.

Insert Table 5 here

Using 100 returns from these strategies we use a t-statistic to test if the average return is statistically different from zero. The trading strategies produce high positive average daily returns for small values of the inertia parameter k (up to 6 basis points for the 2003 samples, up to 4 basis points for the 31 March 2008 sample and only for $k = 0$ for the 1 April 2008 sample). The t-test indicates that the average returns are significantly different from 0 in these cases. Profitability disappears as the frequency of trading decreases. In fact, after $k = 12$, the trading rules start generating negative out-of-sample returns during 13-17 January 2003, for $k = 16$ during 10-14 February 2003 and for $k = 8$ during 17-21 March 2003. For the 2008 samples negative out-of-sample returns start to appear at $k = 6$ for the 31 March sample and at $k = 2$ for the 1 April sample.

Table 6 provides the results for the performance of the “Majority” rule which exhibits similar patterns.

Insert Table 6 here

Out-of-sample returns are positive for small values of k and decrease for lower trading frequencies. Since the “Majority” rule is a combined strategy producing a unique time series of returns we can apply the Anatolyev-Gerko test to examine its predictive ability. For small values of the inertia parameter k (in the range from 0 to 6 with a few exceptions) the “Majority” rule significantly outperforms the random walk strategy for all four sets of conditioning information across the 2003 samples. For 2008 the same applies for k ranging from 0 and 2 basis points for 31 March and only at $k = 0$ for 1 April.

The existence of profitability and the magnitude of returns especially at $k = 0$ are again not very surprising in the absence of transaction costs. One reason for large returns could be simply a reflection of the bid-ask bounce. It is most pronounced at the highest frequency and weakens as the frequency of trading decreases. A more interesting question is whether the profitability of the trading strategies remains when transaction costs are taken into account.

4.1.2 Transaction Costs

Naturally, returns drop considerably when transaction costs are taken into account. Table 7 reports the performance of the linear model when trading on the best bid and ask prices.

Insert Table 7 here

We can see that returns drop substantially when trading at a high frequency. There is no profitability for small values of k , which is to be expected given the higher number of trades. Small values of the inertia parameter reflect a higher number of transactions which implies a large cumulative transaction cost which exceeds the profits from trading. As the number of transactions drops, transaction expenses decrease and trading rules become profitable again.⁶ Out-of-sample returns are positive at k ranges from 6 to 16 basis points during 13-17 January 2003, at k from 8 to 28 during 10-14 February 2003 and k from 10 to 22 during 17-21 March 2003. There are less pronounced patterns of positive returns for the 2008 sample. On 31 March only k values of 10, 20 and 22 generate positive returns and we observe positive returns on 1 April for k ranging from 8 to 12 and from 16 to 20. However, it is important to note that for all samples at $k = 0$ and $k = 2$ the linear trading rule is superior to the random walk across all information sets. Although it generates negative returns, the results indicate that the random walk model loses even more based on trading with transaction costs. For each sample there are inertia values within the 10-14 range which generate positive out-of-sample returns superior to the random walk model (with the exception for the 31 March 2008 sample for *Full* information set).

The highest daily returns are 0.72% during 13-17 January 2003 achieved at $k = 12$, 1.79% during 10-14 February 2003 at $k = 14$, 0.89% during 17-21 March 2003 at $k = 20$, 1.93% during 31 March 2008 and 2.46% on 1 April at $k = 12$. Although these returns may look very high, there is no obvious pattern as to how to exploit them systematically due to the changing nature of the optimal inertia parameter. We address this question at the end of the section. Also, note that traders cannot invest any desired amount of capital into the trading strategy due to our restriction on the trade size. If the trader wanted to invest more than 1 million pounds, he would have to account for the price impact of trades which can substantially reduce profitability of trading strategies.

⁶This result is in line with findings of Knez and Ready (1996), Cooper (1999) and Balvers and Wu (2010) who found that the after-transaction-cost returns with “filter rules” improve compared to trading strategies with zero filter.

Importantly, Table 8 shows that the genetic algorithm can handle transaction costs surprisingly well.

Insert Table 8 here

While the inertia parameter is the only way to mediate the trading frequency for the linear trading rule, the genetic algorithm can adjust it endogenously.⁷ The trading strategy based on the genetic algorithm has positive and significant returns under transaction costs even for low k bands. Profitability vanishes though for values k higher than 12 for the 2003 samples. The signs of average returns during 2008 clearly indicate the lack of a systematic pattern though. Positive returns are generated across all information sets for k equal to 0, 16, 18 on 31 March and only for $k = 10$ on 1 April. Most of the average out-of-sample returns are statistically significant as indicated by the t-test.

Similar results are obtained for the trading strategy based on the “Majority” rule (see Table 9).

Insert Table 9 here

We observe positive returns for the k range between 0 and 10 basis points for the 2003 samples. Most of the strategies exhibit superior performance to the random walk according to the Anatolyev-Gerko test. Returns across the 2008 samples are not systematically positive and change sign from one information set to another.

There is therefore a pronounced difference in the performance of the linear model strategy and the genetic algorithm based strategy under transaction costs. This can be explained by the fact that the genetic algorithm chooses the best in-sample strategy according to its returns taking into account transaction costs. In this way the genetic algorithm based strategy automatically adjusts its trading frequency. In other words, trading rules which tend to trade too often cannot survive in-sample. The linear model strategy cannot adjust to the trading frequency in-sample and therefore it does poorly for small k but improves performance for larger k .

4.1.3 Endogenous inertia parameter

The results reported above show that the profitability of the trading rules critically depends on the value of the inertia parameter. Moreover, the optimal value of k does not stay the same over different sample periods and across different information sets.

⁷Note that genetic algorithm produces different trading rules for trading with no transaction costs and under transaction costs.

Therefore it is important to verify profitability of the trading rules based on an ex-ante and systematic method for the selection of the inertia parameter. In order to do this we endogenize k in the following way. For each value of the inertia parameter the genetic algorithm searches for the best in-sample trading rule. In-sample returns are compared with each other, the rule with the highest return and its corresponding value of k are used to trade out-of-sample. Note, this procedure is using information known to trader at the time of decision making.⁸

The results are provided in Table 10.

Insert Table 10 here

Average out-of-sample returns remain positive across all information sets for the first two samples. During 17-21 March 2003, trading strategies based on *Screen* and limit order book information generate positive and statistically significant returns while the other two information sets fail to generate positive profit. The samples from 2008 do not produce positive returns for any of the information sets.

Up to this point we have not discussed the risk associated with the returns following from each strategy and information set. In Table 10 we present values of Omega measure of performance (see Shadwick and Keating (2002)) along with Sharpe Ratios. Omega is a risk adjusted performance measure, like the Sharpe Ratio, in the sense that it is a ratio of probability weighted gains to losses about a pre-specified threshold τ which we take to be zero,

$$\Omega_{\tau} = \frac{\int_{-\infty}^{\infty} (1 - F(x)) dx}{\int_{-\infty}^{\tau} F(x) dx}$$

As such it reflects the shape of the entire return distribution and all higher moments. The table shows that the Sharpe and Omega Ratios confirm our conclusions from the cumulative returns that there is a substantial difference for all information sets between 2003 and 2008. According to the Omega measure, it is clear that a general bias towards positive returns in 2003 has been replaced, in 2008, by a greater weight being found for negative returns. This is also reflected in the Sharpe ratios. Figure 2 presents a histogram of out-of-sample returns of a randomly selected genetic algorithm generated best in-sample trading rule during 13-17 January 2003. The returns are marked to 1 minute period. The graph clearly demonstrates the substantial downside risk associated with the trading rule.

⁸Balvers and Wu (2010) use a dynamic programming framework to design an ex-ante optimal filter that maximizes expected returns net of transaction costs.

Insert Figure 2 here

We now draw together our results from examining Hypothesis 1. First an interesting and important result is the systematic profitability of trading strategies in the FX market for the 2003 data; both the linear model and the genetic algorithm based approaches generate positive and significant out-of-sample returns even after taking transaction costs into account. These returns could be regarded as a compensation for the risk that traders are exposed to when adopting these strategies as shown by the histogram above. The predictable components in the returns could reflect time-variation in expected returns and the degree of predictability consistent with an efficient market. However, standard performance measures may not reflect the full risks associated with the trading rules. Traders could for instance not only be exposed to market risk as measured by the standard deviation of returns but also to different sources of operational risk.

Another explanation for high out-of-sample returns is simply market inefficiency. The Adaptive Market Hypothesis (AMH) of Lo (2004) states that traders make decisions based on their past experience and learn by receiving negative or positive feedback from the outcomes of their decisions. As a consequence, profitable investment strategies may stop generating positive excess returns because they become more widely exploited using new quantitative methods. Neely et al. (1997) found substantial profitability in exchange rate markets during 1974-1995 using the genetic algorithm based strategy at a daily frequency. However, in a later study (see Neely and Weller (2003)), they could not confirm this result using 1996 half-hourly data. We are effectively confirming this line of argument on a micro level with more recent data. We have found that returns drop substantially in the 2008 samples as compared to 2003. We still see positive returns for some combination of inertia parameter and information sets, but when k is endogenized in the trading strategy, we systematically obtain negative performance. Chaboud, Chiquoine, Hjalmarsson and Vega (2009) report that during 2003 there was almost no algorithmic high-frequency trading in the FX market while the fraction of trading volume where at least one of the two counterparties was an algorithmic trader grew up to 60% by the end of 2007. Thus, the profitability of trading exchange rates at the highest frequency appears to have decreased substantially. This observation would be completely consistent with the AMH.

4.2 Hypothesis 2: Relative Performance

We now turn to Hypothesis 2 and look at the relative performance of the four information sets. In order to test the superior forecasting ability of different conditioning

information we compute the t-statistics of differences between the means of the cumulative returns. This gives us an indication of whether the differential information in each set adds value to the predictions made on the basis of the original, the most basic *Screen* information set. Our main conclusion is that we cannot systematically reject the null hypothesis of the superiority of the *Screen* information. In other words, the enhanced information sets do not appear to add significant value.

4.2.1 No Transaction Costs

The results of the tests for the linear model in the case of no-transaction costs are provided in Table 11.

Insert Table 11 here

This table reports values of the Giacomini-White test of conditional superior predictive ability of each of the extended information sets versus the *Screen* information. Highlighted cells contain significant values of the test statistics at a 5% level; dark grey corresponds to those cases where the *Screen* information outperforms the relevant information set, light grey presents the opposite situation. N/A values appear when we cannot distinguish between the two trading rules – since the trading signals are binary, it is likely that at low frequencies two trading rules generate identical trading signals.

In most of the cases we cannot detect any superior predictive ability for any of the information sets. Only at a very high trading frequency ($k = 0$) do we systematically get significant results. However, there is no uniform pattern in the direction of the results. For the 10-14 February 2003, 31 March 2008 and 1 April 2008 samples all three extended information sets appear to dominate the *Screen* information while during the 13-17 January 2003 and 17-21 March 2003 samples the *Screen* information set significantly outperforms the limit order book and order flow information. The additional information provides no significant improvement out-of-sample

For the genetic algorithm based strategies we get the distribution of out-of-sample returns and employ the paired t-test to compare the performance through average return as above. Table 12 contains the values of the t-statistics.

Insert Table 12 here

Screen information is superior at $k = 0$ and generally at $k = 2$. For all other values of the inertia parameter, the dark and light grey shading appears to be equally scattered across the table with no systematic patterns.

In addition to the t-test based results, we also test for conditional superior predictive ability from the information sets for the genetic algorithm based strategies. Table 13 provides the values of the Giacomini-White test statistics for the “Majority” rule.

Insert Table 13 here

Similar to the previous results, *Screen* information outperforms the limit order book information at $k = 0$. For most of other cases we cannot reject the null hypothesis and obtain any significant results. Again, the additional information contained in the enhanced information sets does not appear to be significant out-of-sample.

4.2.2 Transaction Costs

We next test if the results are robust to the inclusion of transaction costs. For most values of the inertia parameter, the extended information sets do not add significant value to the performance out-of-sample. Values of the Giacomini-White test statistics for the linear model and the “Majority” rule test statistics are provided in Tables 14 and 16 respectively.

Insert Tables 14 and 16 here

Again very few values are statistically significant. For the linear rule significant values appear in the high trading frequency range, where returns are mostly negative. There is no clear systematic pattern among those which generate a statistically significant difference in performance between the information sets. In most cases the three information sets based on limit order book variables are not able to significantly outperform the *Screen* information set.

For the average returns of strategies based on the genetic algorithm we observe much more significant values of the t-statistic (see Table 15).

Insert Table 15 here

For low values of the inertia parameter *Screen* information appears to be superior to limit order book and the order flow information (dark grey shading). There is a k range from 18 to 22 basis points where light grey dominates indicating the superior performance of the limit order book information, but this mostly appears when *Screen* information generates negative average returns (with the exception of the 10-14 February 2003 sample period).

These results again suggest that the limit order book information does not appear to carry significant additional information over that included in past price and quantities, that can be systematically profitable out-of-sample.

This result would be in line with a large theoretical literature claiming that informed traders use market orders to exploit their private information. This means that limit orders and the structure of the book does not carry any substantial information that

can be exploited out-of-sample. There are several papers claiming that order book information has statistical *in-sample* explanatory power which is also confirmed in this paper. However, our main argument is that this predictability cannot be systematically transferred into economically significant profit, at least beyond that which is already in the past price information.

5 Conclusions

This paper has examined the predictability and profitability of the GBP/USD exchange rate using limit order book information. We have tested formally the hypothesis of whether the limit order book information can be profitably exploited out-of-sample for five different samples during 2003 and 2008. Two approaches have been employed to build trading strategies: linear regression and the genetic algorithm.

We have shown that there was a high level of profitability during 2003 and trading strategies generated positive and significant out-of-sample returns net of transaction costs. The level of profitability appears to have dropped considerably during the more recent 2008 data. This finding is in line with the Adaptive Market Hypothesis given that there has been a dramatic rise in algorithmic trading activity since 2003. To the best of our knowledge this gain in efficiency has not yet been reported in the literature for the FX markets but would be consistent with the analysis by Hendershott, Jones and Menkveld (2010) on the impact of algorithmic trading for equity markets.

Our second result is that we could not find any systematic evidence that limit order book information can add significant economic value to the out-of-sample performance of the trading strategies. We look at four information sets, the first of which is based purely on past price information and the quantities of best limit orders which is visible on the screens of trading platforms. The other three are based on the limit order book and order flow information in addition to the screen information set. The information contained in the three enhanced information sets seem to be not robust enough to significantly contribute to the profitability of the trading strategies.

Finally, our results seem to demonstrate that the advent of algorithmic trading has had a huge effect on the efficiency of financial markets. The theoretical models reviewed in Section 2 have been developed when humans were taking trading decisions in FX markets. Chaboud et al. (2009) argue that strategies generated by computers are much more correlated among themselves than humans decisions are. This clearly shows the need to develop new models as to how the algorithmic trading rules are being designed and the extent to which limit orders and the structure of the order book are being exploited in these trading schemes.

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Tables

Table 1: Descriptive Statistics on Market Liquidity

The table presents summary statistics on the liquidity of the market for the 5 subsamples used in the analysis. It reports the mean, standard deviation, median, minimum, maximum and first and the third quartiles of best quantities, slopes of bid and ask sides of limit order book, the depth, inter-quote duration and bid-ask spread. Subsamples are: 13-17 January 2003, 10-14 February 2003, 17-21 March 2003, 31 March 2008 and 1 April 2008. Best quantities and depth are measured in millions of pounds sterling, slopes are basis point per 100 million of currency trade, duration is measured in seconds and bid-ask spread is in basis points.

Variable	Mean	Std. Dev	Min	Q1	Median	Q3	Max	Nr.Orders
13-17 January 2003								
Best bid quantity	2.80	2.53	1	1	2	3	65	75,135
Best ask quantity	2.90	7.45	1	1	2	3	814	
Slope of bid side	58.69	77.87	2.89	25.92	40.00	66.67	3300	
Slope of ask side	65.35	66.08	0.25	28.57	50.00	75.00	1600	
Depth of bid side	44.43	18.45	1	29	44	58	131	
Depth of ask side	39.63	25.86	1	23	32	46	863	
Inter-quote duration	2.29	16.75	0.01	0.30	1.00	2.40	665.51	
Bid-ask spread	2.28	2.06	1	1	2	3	106	
10-14 February 2003								
Best bid quantity	3.07	5.64	1	1	2	3	367	98,785
Best ask quantity	3.03	5.94	1	1	2	3	610	
Slope of bid side	55.53	52.43	2.06	25.00	40.00	66.67	1275	
Slope of ask side	66.71	77.11	0.64	27.77	44.44	75.00	1775	
Depth of bid side	48.81	34.46	2	26	39	56	448	
Depth of ask side	34.21	14.30	1	25	33	42	670	
Inter-quote duration	1.82	3.26	0.01	0.30	0.95	2.11	134.63	
Bid-ask spread	2.31	2.01	1	1	2	3	44	
17-21 March 2003								
Best bid quantity	2.76	2.80	1	1	2	3	71	97,559
Best ask quantity	2.84	3.19	1	1	2	3	95	
Slope of bid side	75.54	101.96	2.66	30.76	50.00	83.33	3600	
Slope of ask side	85.03	98.19	3.44	33.33	57.14	100.00	2040	
Depth of bid side	40.17	15.54	1	28	40	51	108	
Depth of ask side	38.22	18.58	5	25	33	48	154	
Inter-quote duration	1.84	3.33	0.01	0.31	0.92	2.22	152.90	
Bid-ask spread	2.73	3.16	1	1	2	3	119	

Continued

Variable	Mean	Std. Dev	Min	Q1	Median	Q3	Max	Nr.Orders
31 March 2008								
Best bid quantity	3.93	4.06	1	2	3	5	94	594,519
Best ask quantity	3.80	4.63	1	2	3	5	311	
Slope of bid side	20.89	14.54	1.90	12.50	17.64	25.00	671.42	
Slope of ask side	21.52	14.31	0.61	13.33	18.18	25.00	500	
Depth of bid side	79.87	27.69	16	61	78	94	220	
Depth of ask side	72.62	33.12	9	48	67	89	450	
Inter-quote duration	0.10	0.32	0.01	0.01	0.03	0.09	44.96	
Bid-ask spread	2.53	1.21	1	2	2	3	16	
1 April 2008								
Best bid quantity	4.00	4.31	1	2	3	5	72	388,259
Best ask quantity	4.01	6.23	1	2	3	5	310	
Slope of bid side	19.52	12.05	2.53	11.76	16.66	25.00	671.42	
Slope of ask side	20.26	13.50	0.61	12.50	16.66	25.00	500	
Depth of bid side	89.56	32.82	16	70	85	100	220	
Depth of ask side	80.64	32.60	9	55	79	98	450	
Inter-quote duration	0.09	0.28	0.01	0.01	0.03	0.08	30.2	
Bid-ask spread	2.49	1.22	1	2	2	3	17	

Table 2: Number of Indicative Trades

This table provides the number of times the mid-quote of the exchange rate goes outside the k -band for different values of k . When the exchange rate crosses the k -band, trading rules re-evaluate positions taken in the exchange rate. Trade occurs when the trading rule requires a change in the direction of the position. k is measured in basis points.

k	13-17 January 2003	10-14 February 2003	17-21 March 2003	31 March 2008	1 April 2008
0	16,889	22,468	19,654	19,263	12,005
2	1,745	2,775	1,366	1,255	744
4	497	877	443	421	221
6	252	425	215	218	108
8	144	251	140	128	71
10	97	166	99	83	39
12	69	117	64	58	35
14	59	83	45	46	23
16	41	65	34	37	17
18	31	48	24	29	12
20	23	35	22	20	11
22	19	31	22	18	13
24	15	25	21	18	7
26	15	23	15	12	6
28	11	15	14	12	4
30	10	15	10	10	4

Figure 1: Example of genetic algorithm trading rule

The figure presents an example of a genetic algorithm based trading rule. It is a binary logical tree with terminals and operations in their nodes. Given the set of input variables, the rule produces true or false values which correspond to a “buy” and a “sell short” signals respectively. The current example presents a trading rule which generates a signal “buy” if the current quantity weighted spread is less than the past ten ticks average returns times the bid-ask spread and the absolute value of difference between the quantity-weighted bid and the best bid price is less then 0.001. Otherwise, the signal is “sell”.

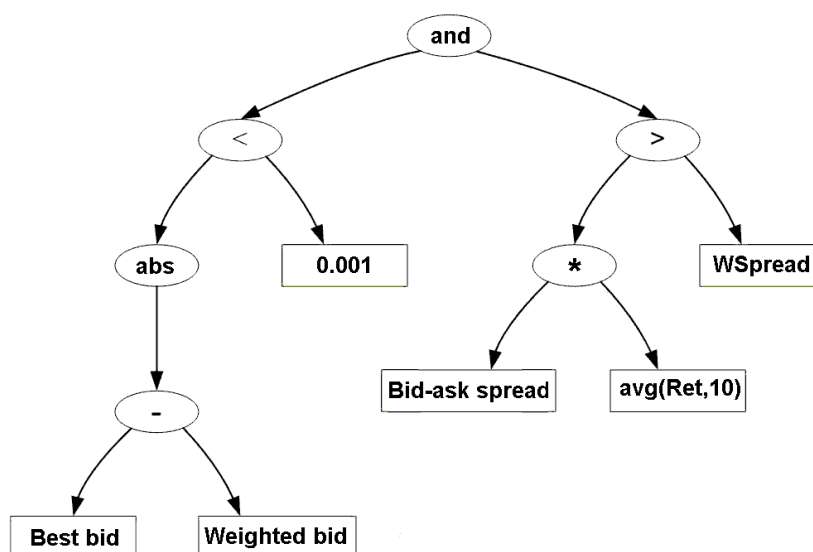


Table 3: Linear Model Parameters Estimates

This table presents coefficient estimates from the linear regression of one step ahead returns on different conditioning variables. The estimation period is the first half of the week commencing on January 13, 2003. The *Screen* column contains estimates of the variables from the *Screen* information set, *Book* corresponds to the limit order book information set, *Order* denotes order flow information and *Full* provides results for the combined information set. * and ** indicate significance at the 5% and 10% levels. The table does not include variables which are not significant in any of the information sets. The F-test row presents F-statistic values for testing the joint significance of all variables added in addition to the *Screen* information set variables (in the *Screen* column the usual F-statistic for significance of the regression is given).

Variable	<i>Screen</i>	<i>Book</i>	<i>Order</i>	<i>Full</i>
Intercept	$-4.43 \cdot 10^{-6**}$	$-4.66 \cdot 10^{-6*}$	$-5.50 \cdot 10^{-6**}$	$-6.93 \cdot 10^{-6**}$
Best ask quantity	$-5.30 \cdot 10^{-7**}$	$-5.28 \cdot 10^{-7**}$	$-5.39 \cdot 10^{-7**}$	$-5.63 \cdot 10^{-7**}$
Best bid quantity	$1.12 \cdot 10^{-6**}$	$9.33 \cdot 10^{-7**}$	$1.01 \cdot 10^{-6**}$	$8.66 \cdot 10^{-7**}$
Ret_{t-1}	$-0.296**$	$-0.295**$	$-0.312**$	$-0.311**$
Inter-quote duration	$5.02 \cdot 10^{-8}$	$7.42 \cdot 10^{-8}$	$9.62 \cdot 10^{-8}$	$1.08 \cdot 10^{-7}$
Bid-ask spread	$0.0116**$	$0.0129**$	$0.0117**$	$0.0129**$
Slope of ask side		$1.08 \cdot 10^{-4**}$		$1.17 \cdot 10^{-4**}$
Slope of bid side		$-2.20 \cdot 10^{-4**}$		$-2.30 \cdot 10^{-4**}$
Depth of ask side		$5.69 \cdot 10^{-8*}$		$5.16 \cdot 10^{-8}$
Mid-quote difference		$0.00121**$		$0.00132**$
Quantity-weighted spread		$2.88 \cdot 10^{-4}$		$2.77 \cdot 10^{-4}$
Best bid order flow(1 tick)			$1.48 \cdot 10^{-6**}$	$1.50 \cdot 10^{-6**}$
Bid cancel order flow(1 tick)			$-8.81 \cdot 10^{-7**}$	$-8.83 \cdot 10^{-7**}$
Bid cancel order flow(1 min)			$8.64 \cdot 10^{-8*}$	$1.08 \cdot 10^{-7**}$
Best ask order flow(1 tick)			$-1.55 \cdot 10^{-6**}$	$-1.51 \cdot 10^{-6**}$
Best ask order flow(20 min)			$2.66 \cdot 10^{-8}$	$5.78 \cdot 10^{-8**}$
Ask order flow(20 min)			$-1.87 \cdot 10^{-8}$	$-4.65 \cdot 10^{-8**}$
Ask cancel order flow(1 tick)			$5.87 \cdot 10^{-7*}$	$6.00 \cdot 10^{-7**}$
Ask cancel order flow(1 min)			$-1.09 \cdot 10^{-7}$	$-1.31 \cdot 10^{-7**}$
Transaction order flow(1 tick)			$1.60 \cdot 10^{-6**}$	$1.43 \cdot 10^{-6**}$
F-test	1176.13	10.96	5.46	7.13

Table 4: Out-of-sample Performance of the Linear Model, No Transaction costs

This table presents the out-of-sample cumulative return generated by the linear model for the five different sample periods. Linear regression coefficients are estimated using in-sample data (first half of the sample) and return predictions are formed for the out-of-sample period (the second half of the sample). A simple binary trading rule is implemented based on the return predictions. Returns are adjusted to a daily basis and expressed in percents. k is the threshold value for the trading band – a trader re-evaluates his position only if the current mid-quote is outside the k -band around the last traded price. k is measured in basis points. There are no transaction costs involved and trading is based solely on mid-quotes returns. Columns contain the results for the different information sets. *Screen* denotes the screen information set, *Book* denotes the limit order book information set, *Order* denotes the order flow set and *Full* corresponds to the combined information set. Bold values indicate positive returns, * and ** indicate significance at the 5% and 1% levels according to Anatolyev-Gerko test statistics.

k	13-17 January 2003				10-14 February 2003				17-21 March 2003				31 March 2008				1 April 2008			
	<i>Screen</i>	<i>Book</i>	<i>Order</i>	<i>Full</i>	<i>Screen</i>	<i>Book</i>	<i>Order</i>	<i>Full</i>	<i>Screen</i>	<i>Book</i>	<i>Order</i>	<i>Full</i>	<i>Screen</i>	<i>Book</i>	<i>Order</i>	<i>Full</i>	<i>Screen</i>	<i>Book</i>	<i>Order</i>	<i>Full</i>
0	12.67**	12.60**	12.56**	12.66**	16.67**	16.92**	16.91**	17.22**	19.97**	19.88**	20.14**	19.96**	29.27**	29.77**	30.88**	33.27**	19.84**	20.52**	21.93**	23.45**
2	4.03**	3.99**	4.04**	4.00**	6.97**	7.10**	6.96**	7.19**	9.68**	9.61**	9.79**	9.73**	0.39	0.43	3.17**	3.06**	3.18	3.14	3.57**	3.60**
4	1.42**	1.33**	1.35**	1.38**	2.00**	2.02**	2.09**	2.12**	3.23**	3.34**	3.25**	3.24**	-0.26	-0.26	1.72	-1.53*	1.27	1.17	1.53	1.56*
6	0.98**	0.97**	0.98**	0.97**	0.84*	0.78**	0.87**	0.78*	0.94*	1.00**	1.07**	1.09**	1.28	1.07	0.63	0.51	0.65	0.65	0.52	-0.23
8	0.57*	0.48	0.46	0.48	0.72*	0.80*	0.63*	0.87*	0.47	0.58	0.45	0.60	1.46	1.53	1.56**	2.29	1.73	1.73	1.42**	1.25
10	0.54	0.54	0.54	0.54	0.85*	0.96**	0.80*	0.87*	0.30	0.36	0.28	0.28	0.99	0.99	0.25	-0.01	1.46	1.46	1.68	1.47
12	0.61*	0.61*	0.49	0.61*	0.99**	0.99**	0.99**	0.99**	0.57	0.57	0.57	0.64	0.50	0.50	1.55	-0.50	2.17	2.17	1.33	0.80
14	0.53*	0.53*	0.53*	0.53*	1.45**	1.37**	1.45**	1.37**	0.83*	0.84*	0.72*	0.72*	0.29	0.29	0.87	-0.55	0.49	0.49	-0.28	-0.28
16	0.22	0.22	0.22	0.22	1.15**	1.23**	1.15**	1.23**	0.37	0.37	0.45	0.44	0.23	0.57	-0.86	-0.41	0.22	0.22	0.30	0.30
18	-0.05	-0.05	0.04	-0.05	1.02**	1.02**	0.93**	0.93**	0.40	0.40	0.31	0.31	1.43	1.43	1.45*	1.81	-0.14	0.24	0.23*	0.23
20	-0.35	-0.35	-0.25	-0.25	0.92**	1.02**	0.92**	1.12**	0.76*	0.86*	0.76*	0.97**	-0.16	-0.16	0.09	-0.12	-0.05	-0.05	-0.27	-0.27
22	-0.02	-0.02	-0.02	-0.02	0.60*	0.60*	0.60*	0.60*	0.77*	0.89*	0.77*	0.89*	0.44	0.44	-0.03	-0.64	0.50	0.50	-0.44	0.04
24	-0.22	-0.22	-0.22	-0.22	0.59*	0.59*	0.47	0.34	0.52	0.52	0.52	0.40	0.33	0.33	-0.17	0.85	-0.21	-0.21	-0.29	-0.29
26	-0.07	-0.07	-0.07	-0.07	0.36	0.36	0.36	0.23	0.05	0.05	0.19	0.19	-0.16	-0.16	-0.15	1.48	-0.87	-0.87	-0.32	-0.32
28	-0.19	-0.19	-0.19	-0.19	0.05	0.05	0.05	0.05	0.30	0.30	0.30	0.30	-0.17	-0.17	-1.32	-0.73	-0.42	-0.42	-0.42	0.15
30	-0.18	-0.18	-0.18	-0.18	0.07	0.25	0.07	0.09	-0.01	0.35	-0.01	0.35	-0.80	-0.80	-0.42	-1.40	-0.50	-0.50	-1.11	-1.11

Table 5: Out-of-sample Performance of the Genetic Algorithm, No Transaction costs

This table presents the out-of-sample cumulative returns (in percents) generated by the genetic algorithm for the five different sample periods. The in-sample period (the first half of the sample) is then used to select the best performing trading rule and this rule is used to trade out-of-sample (the second half of the sample). The exercise is repeated 100 times to generate the empirical distribution of cumulative returns. Returns are adjusted to a daily basis and expressed in percents. k is the threshold value for the trading band – a trader re-evaluates his position only if the current mid-quote is outside the k -band around the last traded price. k is measured in basis points. There are no transaction costs involved and trading is based solely on mid-quotes returns. Columns contain the results for the different information sets. *Screen* denotes the *Screen* information, *Book* denotes the limit order book information set, *Order* denotes the order flow set and *Full* corresponds to the combined information set. Bold values indicate positive returns, * and ** indicate 5% and 1% significance level according to t-test statistics.

k	13-17 January 2003				10-14 February 2003				17-21 March 2003				31 March 2008				1 April 2008			
	<i>Screen</i>	<i>Book</i>	<i>Order</i>	<i>Full</i>	<i>Screen</i>	<i>Book</i>	<i>Order</i>	<i>Full</i>	<i>Screen</i>	<i>Book</i>	<i>Order</i>	<i>Full</i>	<i>Screen</i>	<i>Book</i>	<i>Order</i>	<i>Full</i>	<i>Screen</i>	<i>Book</i>	<i>Order</i>	<i>Full</i>
0	11.01**	10.67**	10.61**	10.13**	14.88**	14.53**	14.43**	13.72**	17.19**	15.68**	15.74**	15.07**	29.01**	26.74**	24.59**	26.98**	19.13**	17.05**	16.56**	17.47**
2	3.69**	3.59**	3.26**	3.22**	5.94**	5.26**	5.25**	5.03**	7.05**	6.53**	6.68**	6.50**	2.09**	2.52**	2.68**	1.65**	1.41**	1.10**	-0.01	-0.36**
4	1.12**	1.22**	1.07**	1.09**	1.38**	1.59**	1.33**	1.53**	0.20	0.67**	0.59**	0.75**	0.07**	0.13	0.51**	0.11	-0.84**	-0.62**	-0.57**	-0.76**
6	0.43**	0.44**	0.01	0.29**	0.43**	0.48**	0.09*	0.34**	0.21**	0.68**	0.11**	0.59**	0.03	-0.03	0.16**	0.05	-0.25**	-0.70**	-0.69**	-0.54**
8	0.07	0.16**	0.14**	0.07	0.66**	0.63**	0.39**	0.55**	0.07	0.32**	-0.20**	0.00	1.46	1.41**	1.06**	0.94	0.08	0.08	0.11	0.05
10	0.15**	0.16**	0.18**	0.21**	0.58**	0.70**	0.36**	0.52**	-0.12**	-0.31**	-0.20**	-0.20**	0.47**	0.26**	0.30**	0.49**	0.33**	0.40**	0.17	0.17
12	-0.04**	0.08*	0.02	0.07	0.45**	0.72**	0.17**	0.51**	-0.11**	-0.08*	0.06*	0.01	0.28**	0.13	0.22*	0.22*	-0.75**	-0.23**	-0.50**	-0.52**
14	-0.12**	0.19**	-0.07	0.15**	0.80**	0.84**	0.62**	0.77**	0.12*	0.08*	0.02	0.15**	0.38**	0.71**	0.03	-0.32**	0.31**	-0.20**	0.10	0.13
16	-0.23**	0.28**	-0.17**	0.14**	-0.09	0.32**	-0.28**	0.08	0.12**	0.01	0.05	-0.05	1.18**	1.34**	0.71**	0.62**	-0.59**	-0.41**	-0.47**	-0.39**
18	-0.04	0.03	-0.14**	0.04	0.20**	0.18**	-0.10*	0.08**	0.15**	0.00	0.13**	0.07	0.44**	0.62**	0.28**	0.48**	-0.47**	-0.36**	0.43**	0.10
20	-0.30**	-0.24**	0.02	-0.10**	-0.07*	0.22**	-0.10*	0.06	0.01	-0.06*	0.02	0.04	-0.99**	-0.70**	-0.68	-0.71**	-0.43**	-0.52**	0.02	0.13
22	-0.15**	-0.16**	-0.06*	-0.09**	0.00	-0.05	0.01	0.05**	0.16**	0.08*	0.09*	0.16**	0.30**	-0.09	-0.23*	-0.43**	0.00	-0.44**	-0.60**	-0.46**
24	-0.11**	-0.12**	-0.01	-0.05	-0.08**	-0.33**	-0.03	-0.25**	0.32**	0.08**	0.06	0.06	-0.99**	-0.14	-0.85	-1.03**	-0.45**	-0.44**	-0.29**	-0.39**
26	-0.18**	-0.17**	-0.04	-0.04	0.03	-0.02	-0.19**	-0.13**	-0.17**	-0.11**	-0.07*	-0.13**	-0.26**	-0.17	-0.23	-0.13	-0.57**	-0.50**	-0.57**	-0.57**
28	-0.01	-0.01	0.06	0.01	-0.04	0.07*	0.04	0.06**	-0.14**	-0.02	-0.01	-0.11**	-0.28**	-0.32**	-0.31	-0.35*	-0.78**	-0.41**	-0.51**	-0.71**
30	-0.09**	-0.10**	-0.06*	-0.08**	0.03	0.02	0.06**	0.06*	-0.16**	-0.18**	-0.12**	-0.18**	-0.06	0.42**	0.08	0.20*	0.04	-0.39*	-0.59**	-0.79**

Table 6: Out-of-sample Performance of the GA “Majority” trading rule, No Transaction costs

This table presents the out-of-sample cumulative returns generated by the “Majority” rule based on the genetic algorithm for the five different sample periods. 99 independent runs of genetic algorithm have been performed to select the best in-sample trading rules. The “Majority” rule produces a “buy” (“sell”) signal if the majority of the 99 best in-sample rules produce a “buy” (“sell”) signal. This combined rule is then used to trade out-of-sample. Returns are adjusted to a daily basis and expressed in percents. k is the threshold value for the trading band – a trader re-evaluates his position only if the current mid-quote is outside the k -band around the last traded price. k is measured in basis points. There are no transaction costs involved and trading is based solely on mid-quotes returns. Columns contain results for the different information sets. *Screen* denotes the screen information, *Book* denotes the limit order book information set, *Order* denotes the order flow set and *Full* corresponds to the combined information set. Bold values indicate positive returns, * and ** indicate significance at the 5% and 1% levels according to Anatolyev-Gerko test statistics.

k	13-17 January 2003				10-14 February 2003				17-21 March 2003				31 March 2008				1 April 2008			
	<i>Screen</i>	<i>Book</i>	<i>Order</i>	<i>Full</i>	<i>Screen</i>	<i>Book</i>	<i>Order</i>	<i>Full</i>	<i>Screen</i>	<i>Book</i>	<i>Order</i>	<i>Full</i>	<i>Screen</i>	<i>Book</i>	<i>Order</i>	<i>Full</i>	<i>Screen</i>	<i>Book</i>	<i>Order</i>	<i>Full</i>
0	10.82**	10.82**	10.82**	10.98**	14.62**	15.69**	16.04**	16.81**	17.46**	16.79**	18.37**	18.82**	34.38**	31.13**	29.94**	34.20**	23.85**	20.56**	21.52**	22.38**
2	3.83**	3.83**	3.81**	3.96**	5.74**	5.94**	6.28**	5.62**	6.86**	7.45**	8.14**	7.99**	2.82**	4.70**	5.16**	4.03**	2.04**	1.72**	-0.65	-0.86
4	1.36**	1.31**	1.40**	1.29**	1.50**	1.81**	1.53**	1.88**	0.29	1.20**	0.93**	1.20**	0.10	0.07	0.61	0.10	-1.49	-1.42	-0.88	-0.88
6	1.07**	0.72*	-0.09	0.60*	0.83**	0.59	0.28	0.83*	0.30	1.13**	0.26	0.83**	-0.06	-0.06	-0.06	-0.06	-0.63	-0.85	-0.34	-0.72
8	-0.11	0.38	0.54*	0.17	0.83*	0.74*	1.33**	0.82*	0.06	0.31*	-0.67**	-0.12	2.29**	1.95*	2.12**	1.77	0.25	0.09	0.06	0.09
10	0.07	0.28	0.21	0.31	0.65*	0.99**	0.75*	0.93**	-0.25	-0.27	-0.31	-0.17	0.82	0.37	0.62	0.81	1.06*	1.74**	0.60	0.61*
12	0.09	-0.28	-0.03	0.15	0.89**	1.20**	0.48	0.95**	-0.10	0.24	0.07	-0.10	1.22	0.66	0.37	-0.44	-1.32	0.53	-1.25	-0.98
14	-0.15	0.28	-0.42	0.24	1.48**	1.33**	1.06**	1.71**	0.31	-0.11	-0.43	-0.12	0.04	0.74	0.47	-0.95	1.49**	-0.42	0.48	0.47
16	-0.37	0.74**	-0.33	0.35	-0.24	0.38	-0.49*	-0.20	0.31*	-0.01	-0.38	-0.55**	1.56	2.57**	0.86	0.37	-1.31	-0.28**	0.05	-0.97
18	-0.01	-0.02	-0.38	0.15	0.21	0.21	-0.23	0.24	0.04	-0.07	0.03	-0.07	0.25	1.57	1.02	0.70	-0.63	-0.71	0.85	0.85
20	-0.63**	-0.21	-0.02	-0.42	-0.14	0.38	-0.22	0.19	-0.22	-0.17	-0.11	-0.32	-1.78	-1.73	-1.78	-1.36	-1.11*	-0.71	0.50	0.49
22	-0.49	-0.26	0.10	0.11	0.09	0.12	-0.09	-0.14	-0.42	0.07	0.08	-0.06	0.90	0.01	0.04	-0.90	0.07	-0.24	-1.16	-1.16
24	-0.02	-0.04	-0.41	-0.03	-0.05	-0.32	-0.05	-0.58	0.37	0.06	-0.14	-0.20	-1.12	0.34	-1.13	-1.13	-0.21	-0.21	-0.21	-0.21
26	0.15*	-0.11	-0.31*	0.03	-0.09	-0.11	-0.37	-0.22	-0.33	0.12	-0.09	-0.20	-0.37	0.36	-0.17	-0.70	-0.71	-0.71	-0.71	-0.73
28	0.01	-0.42*	0.20	0.36	-0.10	0.05	-0.05	-0.05	-0.20	-0.20	-0.20	-0.35	-0.72	-0.40	-0.40	-0.40	-1.28*	-0.42	-0.42	-0.71
30	-0.18	-0.04	0.18	0.02	0.08	0.08	0.20	0.05	-0.20	-0.05	-0.20	-0.32	-0.47	0.82	0.17	0.16	0.11	-0.71	-0.71	-0.71

Table 7: Out-of-sample Performance of the Linear Model, Transaction costs

This table presents the out-of-sample cumulative return generated by the linear model for the five different sample periods. Linear regression coefficients are estimated using in-sample data (first half of the sample) and returns predictions are formed for the out-of-sample period (the second half of the sample). A simple binary trading rule is implemented based on the return predictions. Returns are adjusted to a daily basis and expressed in percents. k is the threshold value for the trading band – a trader re-evaluates his position only if the current mid-quote is outside the k -band around the last traded price. k is measured in basis points. Transaction costs are reflected in bid-ask spread as trading is based on best bid and ask limit orders. Columns contain results for the different information sets. *Screen* denotes the screen information, *Book* denotes the limit order book information set, *Order* denotes the order flow set and *Full* corresponds to the combined information set. Bold values indicate positive returns, * and ** indicate significance at the 5% and 1% levels according to Anatolyev-Gerko test statistics.

	13-17 January 2003			10-14 February 2003			17-21 March 2003			31 March 2008			1 April 2008			
	Screen	Book	Order	Full	Screen	Book	Order	Full	Screen	Book	Order	Full	Screen	Book	Order	Full
0	-37.1**	-36.9**	-36.9**	-36.7**	-47.4**	-46.7**	-47.5**	-46.3**	-53.7**	-53.5**	-53.8**	-53.6**	-80.1**	-80.2**	-79.7**	-78.4**
2	-4.34**	-4.18**	-4.22**	-4.17**	-5.69**	-5.68**	-5.75**	-5.52**	-10.9**	-10.9**	-10.9**	-11.0**	-6.48**	-6.50**	-6.09**	-5.97**
4	-0.10	-0.17	-0.13	-0.18	-0.78	-0.87*	-0.73	-0.74	-2.37**	-2.39**	-2.31**	-2.36**	-1.52**	-1.58**	-1.22	-0.88*
6	0.52	0.56	0.55	0.58	-0.28	-0.36	-0.29	-0.38	-0.80*	-0.73*	-0.66	-0.63	-0.02	-0.02	-0.20**	-1.32
8	0.31	0.25	0.19	0.22	0.41	0.52	0.23	0.46	-0.35	-0.28	-0.40	-0.27	1.49*	1.49*	1.29	0.63
10	0.55*	0.55*	0.55*	0.60*	0.90**	1.00**	0.85**	0.93**	-0.35	-0.31	-0.38	-0.38	1.84**	1.84**	1.97*	1.78**
12	0.72*	0.72*	0.62*	0.61*	1.20**	1.09**	1.20**	1.09**	0.33	0.31	0.33	0.36	2.46**	2.46**	1.23	0.27*
14	0.71**	0.71**	0.71**	0.71**	1.79**	1.71**	1.79**	1.71**	0.72*	0.73*	0.63*	0.64*	1.16*	1.16*	-0.03	-0.03
16	0.19	0.19	0.19	0.19	1.48**	1.56**	1.48**	1.56**	0.27	0.27	0.34	0.32	0.90	0.90	0.51	0.51
18	-0.02	-0.02	0.07	0.07	1.12**	1.12**	1.04**	1.04**	0.51	0.51	0.42	0.42	0.21*	0.59**	0.53	0.53
20	-0.39	-0.39	-0.29	-0.29	1.43**	1.54**	1.43**	1.65**	0.89**	1.00**	0.91**	1.11**	0.31*	0.31*	0.09	0.03
22	-0.04	-0.04	-0.04	-0.04	0.74**	0.74**	0.74**	0.74**	0.88*	1.01**	0.88*	1.01**	1.25**	1.25**	-0.11	0.39
24	-0.26	-0.26	-0.26	-0.26	0.79**	0.79**	0.66**	0.54**	0.64*	0.64*	0.64*	0.54	0.17**	0.17**	-0.34	-0.34
26	-0.11	-0.11	-0.11	-0.11	0.54*	0.43*	0.54*	0.29	-0.10	-0.10	0.03	0.03	-0.46	-0.46	0.05	0.05
28	-0.25	-0.25	-0.25	-0.25	0.09	0.09	0.09	0.09	0.19	0.19	0.19	0.19	-0.44	-0.44	-0.44	0.09
30	-0.20	-0.20	-0.20	-0.20	-0.01	0.16	0.01	0.04	-0.13	0.23	-0.13	0.23	-0.54	-0.54	-1.14	-1.14

Table 8: Out-of-sample Performance of the Genetic Algorithm, Transaction costs

This table presents the out-of-sample cumulative returns (in percents) generated by the genetic algorithm for the five different sample periods. The in-sample period (the first half of the sample) is then used to select the best performing trading rule and this rule is used to trade out-of-sample (the second half of the sample). The exercise is repeated 100 times to generate the empirical distribution of cumulative returns. Returns are adjusted to a daily basis and expressed in percents. k is the threshold value for the trading band – a trader re-evaluates his position only if the current mid-quote is outside the k -band around the last traded price. k is measured in basis points. Transaction costs are reflected in bid-ask spread as trading is based on best bid and ask limit orders. Columns contain results for the different information sets. *Screen* denotes the *Screen* information, *Book* denotes the limit order book information set, *Order* denotes the order flow set and *Full* corresponds to the combined information set. Bold values indicate positive returns, * and ** indicate significance at the 5% and 1% levels according to t-test statistics.

k	13-17 January 2003				10-14 February 2003				17-21 March 2003				31 March 2008				1 April 2008			
	<i>Screen</i>	<i>Book</i>	<i>Order</i>	<i>Full</i>	<i>Screen</i>	<i>Book</i>	<i>Order</i>	<i>Full</i>	<i>Screen</i>	<i>Book</i>	<i>Order</i>	<i>Full</i>	<i>Screen</i>	<i>Book</i>	<i>Order</i>	<i>Full</i>	<i>Screen</i>	<i>Book</i>	<i>Order</i>	<i>Full</i>
0	1.62**	1.60**	1.55**	1.51**	0.96**	0.97**	0.75**	0.78**	0.28**	0.06	-0.12**	-0.10**	0.88**	0.52**	0.45**	0.32**	0.46**	0.07	-0.45**	-0.21**
2	1.50**	1.50**	1.41**	1.43**	0.97**	0.96**	0.94**	0.87**	0.39**	0.22**	0.08*	-0.07*	0.04	-0.12	-0.07	-0.05	-0.74**	-0.71**	-1.99**	-2.01**
4	1.57**	1.39**	1.26**	1.30**	1.10**	1.12**	0.89**	0.84**	0.42**	0.31**	0.00	-0.01	0.26**	-0.83**	-0.50**	-0.24**	-0.85**	-0.68**	-1.26**	-1.15**
6	0.87**	0.42**	0.20*	0.07	0.93**	0.72**	0.51**	0.19**	0.22**	0.09*	-0.11*	-0.18**	-2.00**	-1.69**	-1.53*	-1.86**	-1.42**	-1.31**	-1.19**	-1.15**
8	0.83**	0.61**	0.57**	0.45**	0.78**	0.61**	0.40**	0.38**	0.03	0.02	-0.36**	-0.34**	0.50**	0.41**	0.05	0.33**	-0.26**	-0.17	-0.49**	-0.75**
10	0.59**	0.41**	0.59**	0.41**	0.82**	0.76**	0.60**	0.77**	-0.17**	-0.14**	-0.28**	-0.31**	-0.04	-0.04	-0.30**	-0.09	0.28**	0.06	0.37**	0.30**
12	0.13*	0.34**	0.11	0.31**	1.03**	1.02**	0.58**	0.73**	0.11*	0.00	-0.01	-0.05	-0.08	0.06	0.07	0.17	-0.49**	-0.12	-0.67**	-0.60**
14	0.21**	0.48**	-0.01	0.25**	1.11**	1.13**	0.86**	1.08**	0.21**	0.15**	0.22**	0.19**	0.40**	0.55**	-0.10	-0.49**	0.37**	-0.10	0.13	0.41**
16	0.04	0.45**	-0.07	0.13*	0.51**	0.56**	-0.12**	0.31**	0.08*	0.05*	-0.04	-0.13**	1.29**	0.87**	0.51**	0.74**	-0.25**	0.04	-0.01	-0.33**
18	-0.03	0.13**	-0.10*	-0.05	0.28**	0.23**	-0.21**	-0.05	0.19**	0.04	0.25**	0.17**	0.40**	0.72**	0.27**	0.42**	-0.44**	-0.31**	0.12	0.21**
20	-0.10**	-0.09*	0.08*	0.12**	0.25**	0.39**	0.34**	0.33**	-0.08*	-0.10**	0.19**	0.10*	-1.13**	-0.61**	-0.79**	-1.15**	-0.42**	-0.33**	0.08	0.29**
22	0.00	-0.09**	-0.03	-0.07*	-0.08*	0.10**	-0.02	0.02	0.19**	0.24**	0.18**	0.24**	0.13	-0.21*	-0.07	0.01	0.53**	-0.11	-0.24**	-0.06
24	-0.02	-0.06	0.06*	0.00	0.06	-0.35**	0.08*	-0.10*	0.39**	0.05	0.26**	0.16**	-0.99**	-0.53**	-1.28*	-1.32**	-0.03**	-0.08	-0.21**	-0.13**
26	-0.14**	-0.10**	-0.04	-0.04	0.19**	0.06	-0.14**	-0.08*	-0.10*	-0.24**	-0.16**	-0.14**	-0.11	-0.16**	-0.24**	0.05	-0.16	-0.25**	-0.26**	-0.37**
28	0.01	-0.12**	0.03	-0.06	0.05	-0.04	0.07**	0.01	-0.03	-0.02	-0.04	-0.07*	-0.27**	-0.29**	-0.46**	-0.40**	-0.80**	-0.47**	-0.53**	-0.71**
30	-0.05	-0.03	-0.02	-0.03	-0.01	-0.04	0.00	-0.06*	-0.22**	-0.33**	-0.18**	-0.25**	-0.07	0.42**	0.29**	0.12	-0.10**	-0.47**	-0.70**	-0.69**

Table 9: Performance of the GA “Majority” trading rule, Transaction costs

This table presents the out-of-sample cumulative returns generated by the “Majority” rule based on genetic algorithm for the five different sample periods. 99 independent runs of the genetic algorithm have been performed to select the best in-sample trading rules. The “Majority” rule produces a “buy” (“sell”) signal if the majority of the 99 best in-sample rules produce a “buy” (“sell”) signal. This combined rule is then used to trade out-of-sample. Returns are adjusted to a daily basis and expressed in percents. k is the threshold value for the trading band – a trader re-evaluates his position only if the current mid-quote is outside the k -band around the last traded price. k is measured in basis points. Transaction costs are reflected in bid-ask spread as trading is based on best bid and ask limit orders. Columns contain results for the different information sets. *Screen* denotes the screen information, *Book* denotes the limit order book information set, *Order* denotes the order flow set and *Full* corresponds to the combined information set. Bold values indicate positive returns, * and ** indicate significance at the 5% and 1% levels according to Anatolyev-Gerko test statistics.

k	13-17 January 2003				10-14 February 2003				17-21 March 2003				31 March 2008				1 April 2008			
	<i>Screen</i>	<i>Book</i>	<i>Order</i>	<i>Full</i>	<i>Screen</i>	<i>Book</i>	<i>Order</i>	<i>Full</i>	<i>Screen</i>	<i>Book</i>	<i>Order</i>	<i>Full</i>	<i>Screen</i>	<i>Book</i>	<i>Order</i>	<i>Full</i>	<i>Screen</i>	<i>Book</i>	<i>Order</i>	<i>Full</i>
0	1.98**	1.93**	2.23**	2.32**	1.29**	1.30**	0.83**	0.88**	0.67**	0.59**	0.43**	0.54**	0.13	0.13	0.07	0.13	0.23	-0.72	-0.72	-0.72
2	1.73**	1.79**	1.74**	1.71**	1.14**	1.26**	1.26**	1.13**	0.56**	0.59**	0.51**	-0.09	0.07	0.07	0.07	0.07	-0.72	-0.72	-1.87	-2.01
4	1.60**	1.66**	1.61**	1.64**	1.34**	1.39**	1.29**	1.30**	0.65**	0.57**	0.57**	0.54**	0.88**	-1.42	-2.10*	0.37	-0.82**	-0.72	-1.08*	-1.08
6	1.64**	0.63	0.54	-0.05	1.28**	1.39**	0.99**	1.19**	0.52**	0.54**	0.52**	0.36**	-2.25**	-2.43**	-2.49**	-2.43**	-1.81**	-1.53**	-1.39**	-1.39**
8	1.73**	1.35**	1.10**	1.33**	1.06**	0.93**	0.77**	0.30	0.36**	0.65**	0.04*	0.33**	0.51	0.58	1.09	0.10	-0.96	0.17	-0.46	-1.40
10	0.69*	0.41	0.87*	0.91**	0.89**	1.08**	0.80**	1.15**	0.24**	0.45**	0.16**	-0.23**	0.02	0.45	0.18	-0.04	0.53	-0.64	1.10	0.90
12	-0.16	0.32	0.18	0.75*	0.81**	1.84**	0.88**	0.65*	0.39**	0.21	-0.08	0.16	0.27	-0.12	-0.61	-0.33	-0.48	0.08	-1.46	-1.08
14	0.49	0.37	0.00	0.77*	1.12**	1.53**	1.34**	1.92**	0.33	0.15**	0.22	0.10	0.82	1.01	0.42	-1.58	1.45	-0.48	0.09	1.20
16	-0.16	0.92**	-0.49	-0.17	0.38	1.09**	-0.64**	0.75**	0.19	0.13**	-0.12	-0.59*	2.57**	2.57**	0.25	0.59	-0.04**	-0.01**	0.35	-0.09
18	0.05	0.22	-0.03	0.02	0.53*	0.55*	-0.36*	-0.16	0.14	-0.04	0.07	0.20**	0.80	1.65	1.30	0.76	0.45	-0.72	1.14	1.14
20	-0.32	-0.28	-0.06	0.61	0.44*	0.74**	0.34	-0.06	-0.32	-0.19**	-0.08	-0.11	-1.58	-1.18	-2.00**	-2.44**	-1.14	-0.72	0.39**	0.39**
22	-0.01	-0.39	-0.15	-0.29	-0.03	-0.02	0.09	0.04	0.04	0.13	0.29	0.11	0.60	0.29	-1.67*	0.13	0.43	-1.02	-0.70*	-1.20
24	0.02	-0.05	0.07	0.14	0.12	-0.55*	0.12	-0.35	0.55**	0.11	0.03	-0.15	-1.19	-0.81	-1.45	-1.45	0.17	0.17	0.17	0.17
26	-0.19	0.16	-0.19*	-0.19*	0.28*	-0.10	-0.41*	-0.43	-0.34	-0.31	-0.04*	-0.24	-0.29	-0.29	0.00	-0.29	0.33	-0.76	-0.20	-0.38
28	0.08	-0.19	-0.01*	-0.15*	0.22	-0.04	-0.01	0.04	0.06*	-0.22	-0.22	-0.22	-0.40	-0.40	-0.40	-0.40	-1.30	-0.44	-0.44	-0.72
30	-0.12	0.17	0.07	-0.02	-0.11	0.01	-0.01	0.20	-0.23	-0.10	-0.21	-0.08	0.02	0.71	0.08	0.08	0.03	-0.72	-0.72	-0.72

Table 10: Endogenous k , Genetic Algorithm Performance

This table presents the out-of-sample cumulative returns generated by the genetic algorithm for the five difference sample periods when k is optimally determined based on the in-sample performance. k reflects the inertia threshold value, i.e. the trader re-evaluates his position only if the current mid-quote is outside the k -band around the last traded price. The in-sample period is used to select the best performing trading rule for each k . The best performing rule at k generating the highest in-sample return is used to trade out-of-sample. The exercise is repeated 100 times to generate the empirical distribution of cumulative returns. Returns are adjusted to a daily basis. Transaction costs are reflected in bid-ask spread. Average GA returns column provides average out-of-sample returns, Ω contains the Omega measure, the last panel presents Sharpe Ratios of the trading rules. *Screen* denotes the Screen information, *Book* denotes the limit order book information set, *Order* denotes the order flow set and *Full* corresponds to the combined information set. Bold values indicate positive returns, * and ** indicate significance at the 5% and 1% levels according to t-test statistics.

Sample	Average GA returns				Ω				Sharpe Ratio			
	<i>Screen</i>	<i>Book</i>	<i>Order</i>	<i>Full</i>	<i>Screen</i>	<i>Book</i>	<i>Order</i>	<i>Full</i>	<i>Screen</i>	<i>Book</i>	<i>Order</i>	<i>Full</i>
13-17 Jan 2003	0.65**	0.57**	0.57**	0.44**	34.9	23.9	15.9	12.9	1.67	1.35	1.31	1.19
10-14 Feb 2003	0.98**	0.94**	0.66**	0.66**	96.5	80.2	28.5	53.2	2.81	1.99	1.31	1.28
17-21 Mar 2003	0.15**	0.10**	-0.05	-0.04**	2.48	1.73	0.73	0.74	0.39	0.22	-0.11	-0.12
31 Mar 2008	-0.65**	-0.09**	-0.51**	-0.68**	0.21	0.82	0.35	0.21	-0.59	-0.08	-0.42	-0.20
1 Apr 2008	-0.06	-0.07	-0.15**	-0.18	0.81	0.80	0.56	0.60	-0.08	-0.09	-0.22	-0.63

Figure 2: Histogram of Returns of a Genetic Algorithm Trading Rule

The figure presents the out-of-sample performance of a typical genetic algorithm trading rule at $k = 4$ during 13-17 January 2003 sample. The upper figure presents time series of cumulative returns marked at one minute frequency (black line) and mid-quotes of the exchange rate during the out-of-sample period. The lower figure presents the corresponding histogram of one-minute returns.

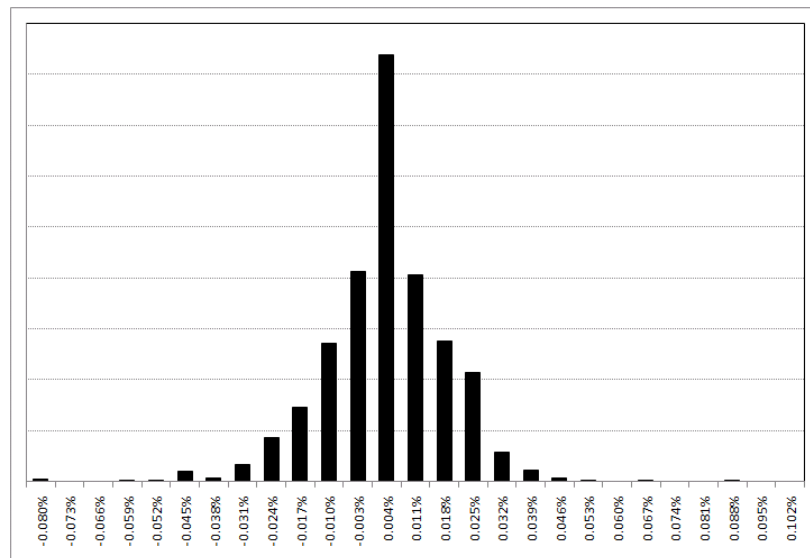


Table 11: Relative Performance of the Linear Model, No Transaction costs

The table presents values of Giancomini-White test statistics for comparing the relative out-of-sample performance of the linear model based on three information sets against the screen information. The linear regression coefficients are estimated using in-sample data and return predictions are formed for the out-of-sample period. A simple binary trading rule is implemented based on the return predictions. There are no transaction costs involved and trading is based solely on mid-quotes returns. k is the threshold value for the trading band and measured in basis points. Columns $Book - Screen$ contains statistics for relative performance of the limit order book vs. the screen information, $Order - Screen$ compares the order flow vs. the screen information and $Full - Screen$ corresponds to the combined information set vs. the screen information. Highlighted cells contain significant values at 5% level; dark grey colour corresponds to the cases where the *Screen* information outperforms the corresponding information set, light grey colour presents the opposite situation.

	13-17 January 2003			10-14 February 2003			17-21 March 2003			31 March 2008			1 April 2008		
k	$Book - Screen$	$Order - Screen$	$Full - Screen$	$Book - Screen$	$Order - Screen$	$Full - Screen$	$Book - Screen$	$Order - Screen$	$Full - Screen$	$Book - Screen$	$Order - Screen$	$Full - Screen$	$Book - Screen$	$Order - Screen$	$Full - Screen$
0	62.76	32.55	147.9	16.44	17.0	15.38	14.55	44.61	8.63	387.5	47.52	15.18	516.8	51.50	30.47
0.0002	1.02	4.47	0.68	1.05	8.57	2.09	0.85	2.11	0.11	2.19	13.82	7.40	3.53	0.79	1.14
0.0004	6.32	2.44	3.06	2.18	4.20	3.91	1.69	1.11	2.75	0.12	2.40	3.65	40.7	0.65	0.92
0.0006	4.08	0.88	4.08	0.60	2.74	0.12	0.74	1.61	2.11	2.38	1.18	2.52	1.35	1.97	0.92
0.0008	2.28	1.60	2.28	0.31	1.63	1.14	1.48	1.46	2.07	N/A	0.68	0.28	N/A	1.17	2.01
0.001	N/A	N/A	N/A	1.20	1.47	0.01	6.93	1.22	0.92	N/A	2.05	0.47	N/A	0.06	0.51
0.0012	N/A	1.97	N/A	2.12	N/A	2.12	0.06	N/A	1.58	N/A	1.60	1.04	N/A	2.04	2.00
0.0014	N/A	N/A	N/A	2.13	N/A	2.13	1.82	2.72	4.90	N/A	0.87	0.79	N/A	2.44	2.44
0.0016	N/A	N/A	N/A	1.80	N/A	1.80	N/A	2.03	1.03	N/A	2.24	1.28	N/A	3.01	3.01
0.0018	N/A	1.96	N/A	N/A	2.06	2.06	N/A	1.03	1.03	N/A	0.95	1.69	N/A	2.14	2.14
0.002	N/A	2.00	2.00	2.15	N/A	2.00	1.97	0.00	3.61	N/A	3.51	4.50	N/A	0.00	0.00
0.0022	N/A	N/A	N/A	N/A	N/A	N/A	2.03	N/A	2.03	N/A	1.21	2.02	N/A	0.77	2.00
0.0024	N/A	N/A	N/A	N/A	2.04	3.74	N/A	N/A	2.03	N/A	0.98	0.38	N/A	2.00	2.00
0.0026	N/A	N/A	N/A	0.10	N/A	1.79	N/A	1.98	1.98	N/A	2.99	1.17	N/A	2.00	2.00
0.0028	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	2.00	1.93	N/A	2.00	N/A
0.003	N/A	N/A	N/A	1.98	N/A	0.78	2.93	N/A	2.93	N/A	1.99	0.00	N/A	N/A	N/A

Table 12: Relative Performance of the Genetic Algorithm Model, No Transaction costs

The table presents values of paired t-test statistics for comparing the relative out-of-sample performance of the genetic algorithm model based on three information sets against the screen information. The in-sample period (the first half of the sample) is used to select the best performing trading rule and this rule is used to trade out-of-sample (the second half of the sample). The exercise is repeated 100 times to generate the empirical distribution of cumulative returns. There are no transaction costs involved and trading is based solely on mid-quotes returns. k is the threshold value for the trading band and measured in basis points. Columns *Book – Screen* contains statistics for relative performance of the limit order book vs. the screen information, *Order – Screen* compares the order flow vs. the screen information and *Full – Screen* corresponds to the combined information set vs. the screen information. Highlighted cells contain significant values at 5% level; dark grey colour corresponds to the cases where the *Screen* information outperforms the corresponding information set, light grey colour presents the opposite situation.

	13-17 January 2003			10-14 February 2003			17-21 March 2003			31 March 2008			1 April 2008		
k	<i>Book – Screen</i>	<i>Order – Screen</i>	<i>Full – Screen</i>	<i>Book – Screen</i>	<i>Order – Screen</i>	<i>Full – Screen</i>	<i>Book – Screen</i>	<i>Order – Screen</i>	<i>Full – Screen</i>	<i>Book – Screen</i>	<i>Order – Screen</i>	<i>Full – Screen</i>	<i>Book – Screen</i>	<i>Order – Screen</i>	<i>Full – Screen</i>
0	-1.62	-2.12	-3.46	-1.22	-1.69	-3.49	-3.39	-2.97	-4.19	-3.16	-6.49	-2.75	-4.25	-4.44	-2.97
0.0002	-1.48	-4.21	-4.97	-6.00	-4.85	-7.34	-2.67	-1.88	-2.95	1.94	2.81	-2.10	-1.65	-8.30	-12.69
0.0004	2.20	-1.02	-0.62	3.75	-0.96	2.71	2.29	2.02	2.94	0.49	3.21	0.25	1.53	2.24	0.75
0.0006	0.14	-5.60	-1.58	0.93	-4.79	-1.44	5.71	-1.54	4.02	-0.79	1.50	0.32	-3.20	-3.39	-2.21
0.0008	1.19	0.97	-0.01	-0.61	-4.65	-2.16	2.95	-3.18	-0.76	-0.35	-2.18	-3.08	-0.02	0.32	-0.25
0.001	0.09	0.36	0.73	2.00	-3.32	-0.98	-3.17	-1.25	-1.15	-1.37	-1.00	0.14	0.47	-1.03	-0.93
0.0012	1.88	0.92	1.62	3.72	-2.90	0.84	0.58	3.27	2.02	-0.81	-0.37	-0.31	3.11	1.32	1.30
0.0014	3.84	0.57	3.53	0.54	-2.14	-0.35	-0.51	-1.45	0.53	2.19	-2.12	-4.02	-3.78	-1.57	-1.48
0.0016	8.53	1.12	5.84	4.94	-2.20	1.85	-2.01	-1.42	-2.86	0.99	-3.15	-3.48	1.94	1.29	2.07
0.0018	1.11	-1.66	1.17	-0.39	-5.17	-2.14	-2.94	-0.35	-1.30	1.15	-0.91	0.21	1.27	9.96	6.43
0.002	1.33	6.07	4.06	4.84	-0.54	2.28	-1.29	0.25	0.61	1.93	1.70	1.61	-0.71	4.54	5.93
0.0022	-0.13	1.69	1.40	-0.90	0.21	1.00	-1.29	-1.44	-0.06	-2.25	-3.50	-5.40	-2.98	-4.05	-3.06
0.0024	-0.20	1.71	1.12	-5.54	0.72	-3.31	-4.47	-4.74	-4.58	6.29	1.06	-0.29	0.09	1.56	0.52
0.0026	0.13	3.15	3.02	-0.76	-3.71	-2.70	0.85	1.55	0.65	0.59	0.15	0.91	0.79	0.01	-0.01
0.0028	-0.05	1.25	0.33	2.45	1.84	2.40	2.10	2.64	0.52	-0.35	-0.25	-0.49	5.97	4.33	1.02
0.003	-0.14	0.57	0.14	-0.27	0.66	0.56	-0.63	1.00	-0.42	2.96	0.83	1.61	-4.91	-7.62	-10.58

Table 13: Relative Performance of the GA “Majority” trading rule, No Transaction costs

The table presents values of Giancomini-White test statistics for comparing the relative out-of-sample performance of the “Majority” rule based on three information sets against the screen information. 99 independent runs of the genetic algorithm have been performed to select the best in-sample trading rules. The “Majority” rule produces a “buy” (“sell”) signal if the majority of the 99 best in-sample rules produce a “buy” (“sell”) signal. This combined rule is then used to trade out-of-sample. There are no transaction costs involved and trading is based solely on mid-quotes returns. k is the threshold value for the trading band and measured in basis points. Columns *Book – Screen* contains statistics for relative performance of the limit order book vs. the screen information, *Order – Screen* compares the order flow vs. the screen information and *Full – Screen* corresponds to the combined information set vs. the screen information. Highlighted cells contain significant values at 5% level; dark grey colour corresponds to the cases where the *Screen* information outperforms the corresponding information set, light grey colour presents the opposite situation.

	13-17 January 2003			10-14 February 2003			17-21 March 2003			31 March 2008			1 April 2008		
k	<i>Book – Screen</i>	<i>Order – Screen</i>	<i>Full – Screen</i>	<i>Book – Screen</i>	<i>Order – Screen</i>	<i>Full – Screen</i>	<i>Book – Screen</i>	<i>Order – Screen</i>	<i>Full – Screen</i>	<i>Book – Screen</i>	<i>Order – Screen</i>	<i>Full – Screen</i>	<i>Book – Screen</i>	<i>Order – Screen</i>	<i>Full – Screen</i>
0	2833.8	2833.8	764.82	3.09	1.78	1.95	28.65	5.22	6.02	25.02	16.60	20.41	16.63	18.35	17.91
0.0002	1.32	3.31	18.53	1.13	0.06	0.19	12.45	29.17	26.46	3.27	3.98	1.73	1.46	10.55	10.72
0.0004	49.41	0.94	5.62	1.60	0.00	0.79	5.39	2.69	3.71	0.07	0.56	2.01	2.01	3.15	3.16
0.0006	3.31	7.72	4.77	5.96	8.52	10.13	7.87	1.93	3.35	N/A	N/A	N/A	2.42	1.82	0.09
0.0008	2.41	3.59	4.08	2.43	3.37	2.03	0.89	5.66	0.38	4.58	5.44	0.55	0.98	0.98	0.75
0.001	0.58	2.12	0.90	1.09	0.33	0.36	0.44	0.06	1.32	1.22	1.01	0.33	2.62	0.94	1.07
0.0012	3.57	3.03	2.04	1.74	0.70	3.24	0.93	2.11	1.07	2.73	0.92	2.40	3.32	1.45	0.33
0.0014	3.36	2.09	2.74	0.78	2.89	2.18	1.73	4.40	2.38	3.14	3.23	0.88	4.71	2.79	1.43
0.0016	7.56	0.07	4.11	5.18	3.47	3.07	2.20	6.49	9.98	3.00	1.32	1.84	1.35	2.34	1.00
0.0018	0.21	4.00	0.28	1.41	0.82	0.52	0.29	0.20	0.29	1.28	4.91	4.70	0.03	3.85	3.85
0.002	4.36	3.12	1.38	0.90	1.38	2.35	4.01	2.94	0.19	1.40	0.00	1.37	N/A	4.00	2.72
0.0022	0.73	2.28	4.00	2.41	2.86	1.42	2.15	1.28	3.01	2.01	2.11	1.56	0.63	1.34	1.34
0.0024	3.63	1.16	1.89	0.33	1.67	0.01	1.23	2.04	3.99	3.87	2.63	2.63	N/A	N/A	N/A
0.0026	1.15	1.67	1.01	5.05	0.01	0.01	3.65	1.04	2.00	1.26	1.98	1.31	N/A	N/A	0.00
0.0028	3.85	0.45	2.14	1.33	0.52	1.66	N/A	2.79	2.00	N/A	N/A	N/A	N/A	N/A	N/A
0.003	0.32	2.72	2.72	0.69	3.13	4.02	1.06	N/A	2.00	2.00	2.00	1.00	2.00	2.00	2.00

Table 14: Relative Performance of the Linear Model, Transaction costs

The table presents values of Giancomini-White test statistics for comparing the relative out-of-sample performance of the linear model based on three information sets against the screen information. The linear regression coefficients are estimated using in-sample and return predictions are formed for the out-of-sample period. A simple binary trading rule is implemented based on the return predictions. Transaction costs are reflected in the bid-ask spread as trading is based on best bid and ask limit orders. k is the threshold value for the trading band and measured in basis points. Columns $Book - Screen$ contains statistics for relative performance of the limit order book vs. the screen information, $Order - Screen$ compares the order flow vs. the screen information and $Full - Screen$ corresponds to the combined information set vs. the screen information. Highlighted cells contain significant values at 5% level; dark grey colour corresponds to the cases where the *Screen* information outperforms the corresponding information set, light grey colour presents the opposite situation.

	13-17 January 2003			10-14 February 2003			17-21 March 2003			31 March 2008			1 April 2008		
k	$Book - Screen$	$Order - Screen$	$Full - Screen$	$Book - Screen$	$Order - Screen$	$Full - Screen$	$Book - Screen$	$Order - Screen$	$Full - Screen$	$Book - Screen$	$Order - Screen$	$Full - Screen$	$Book - Screen$	$Order - Screen$	$Full - Screen$
0	5.79	2.20	7.29	16.70	1.71	22.02	6.72	41.71	2.46	5.79	2.20	7.29	10.14	47.01	19.23
0.0002	2.17	4.52	1.58	1.46	3.33	1.57	1.46	5.62	6.17	2.17	4.52	1.58	0.02	4.09	6.95
0.0004	6.04	1.54	4.77	1.01	3.42	0.39	0.59	1.46	1.78	6.04	1.54	4.77	2.92	3.39	0.47
0.0006	2.03	0.13	0.45	6.38	0.38	4.13	0.50	1.89	1.89	2.03	0.13	0.45	N/A	2.62	1.78
0.0008	1.19	3.39	1.92	2.01	2.66	0.51	0.69	0.24	0.74	1.19	3.39	1.92	N/A	1.52	2.55
0.001	N/A	N/A	1.06	0.92	2.75	0.59	0.15	0.13	0.44	N/A	N/A	1.06	N/A	0.04	0.08
0.0012	N/A	1.95	1.46	0.40	N/A	0.40	0.08	N/A	0.13	N/A	1.95	1.46	N/A	5.34	2.49
0.0014	N/A	N/A	N/A	1.59	N/A	1.59	2.34	2.39	3.95	N/A	N/A	N/A	N/A	3.75	3.75
0.0016	N/A	0.82	N/A	0.49	N/A	0.49	N/A	1.09	2.38	N/A	0.82	N/A	N/A	0.77	0.77
0.0018	N/A	2.71	0.82	N/A	2.34	2.34	N/A	2.97	2.97	N/A	2.71	0.82	N/A	0.12	0.12
0.002	N/A	N/A	2.71	0.77	N/A	1.71	2.47	2.02	4.82	N/A	N/A	2.71	N/A	0.06	1.41
0.0022	N/A	N/A	N/A	N/A	N/A	N/A	2.23	N/A	2.23	N/A	N/A	N/A	N/A	0.79	3.07
0.0024	N/A	N/A	N/A	N/A	2.89	3.43	N/A	N/A	2.11	N/A	N/A	N/A	N/A	1.83	1.83
0.0026	N/A	N/A	N/A	1.95	N/A	1.72	N/A	2.00	2.00	N/A	N/A	N/A	N/A	0.14	0.14
0.0028	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	2.00	N/A
0.003	N/A	N/A	N/A	2.99	N/A	1.15	1.07	N/A	1.07	N/A	N/A	N/A	N/A	2.00	2.00

Table 15: Relative Performance of the Genetic Algorithm Model, Transaction costs

The table presents values of paired t-test statistics for comparing the relative out-of-sample performance of the genetic algorithm model based on three information sets against the screen information. The in-sample period (the first half of the sample) is then used to select the best performing trading rule and this rule is used to trade out-of-sample (the second half of the sample). The exercise is repeated 100 times to generate the empirical distribution of cumulative returns. Transaction costs are reflected in the bid-ask spread as trading is based on the best bid and ask limit orders. k is the threshold value for the trading band and measured in basis points. Columns *Book – Screen* contains statistics for relative performance of the limit order book vs. the screen information, *Order – Screen* compares the order flow vs. the screen information and *Full – Screen* corresponds to the combined information set vs. the screen information. Highlighted cells contain significant values at 5% level; dark grey colour corresponds to the cases where the *Screen* information outperforms the corresponding information set, light grey colour presents the opposite situation.

	13-17 January 2003			10-14 February 2003			17-21 March 2003			31 March 2008			1 April 2008		
k	<i>Book – Screen</i>	<i>Order – Screen</i>	<i>Full – Screen</i>	<i>Book – Screen</i>	<i>Order – Screen</i>	<i>Full – Screen</i>	<i>Book – Screen</i>	<i>Order – Screen</i>	<i>Full – Screen</i>	<i>Book – Screen</i>	<i>Order – Screen</i>	<i>Full – Screen</i>	<i>Book – Screen</i>	<i>Order – Screen</i>	<i>Full – Screen</i>
0	-0.46	-1.17	-1.54	0.12	-2.72	-2.09	-3.84	-6.99	-7.61	-1.96	-2.63	-3.58	-2.09	-6.09	-4.20
0.0002	-0.22	-2.37	-1.85	-0.20	-0.55	-1.59	-3.49	-5.58	-9.88	-1.56	-1.82	-1.70	0.51	-22.85	-22.67
0.0004	-2.73	-3.34	-2.95	0.22	-2.82	-3.04	-2.20	-5.21	-6.58	-5.96	-4.50	-3.26	1.71	-4.88	-3.84
0.0006	-3.05	-4.51	-5.88	-2.52	-3.98	-7.71	-2.26	-4.68	-5.92	2.18	2.54	0.99	0.84	1.58	2.66
0.0008	-1.43	-1.85	-2.57	-2.31	-5.01	-6.69	-0.09	-4.43	-4.12	-0.68	-2.30	-1.07	0.60	-1.63	-3.65
0.001	-1.88	-0.03	-1.79	-0.77	-2.98	-0.61	0.44	-1.75	-2.13	-0.02	-1.75	-0.34	-1.39	0.52	0.11
0.0012	2.09	-0.11	1.93	-0.15	-5.35	-3.54	-1.26	-1.66	-2.35	0.78	0.88	1.46	2.52	-1.04	-0.76
0.0014	3.01	-2.20	0.51	0.26	-2.69	-0.45	-0.86	0.14	-0.28	0.98	-2.80	-5.26	-3.03	-1.62	0.27
0.0016	5.18	-1.58	1.12	0.66	-8.66	-2.30	-0.47	-2.16	-3.85	-1.97	-4.11	-2.79	2.98	2.43	-0.86
0.0018	2.55	-1.27	-0.32	-0.72	-8.12	-4.94	-2.70	1.02	-0.31	2.33	-0.82	0.11	1.33	5.22	5.77
0.002	0.11	3.09	3.78	2.36	2.00	1.49	-0.41	3.87	2.68	3.32	2.30	-0.14	0.87	5.39	6.94
0.0022	-1.97	-0.66	-1.57	3.37	1.19	1.92	0.69	-0.23	0.74	-2.11	-1.28	-0.76	-3.65	-4.41	-3.31
0.0024	-0.73	1.73	0.43	-7.24	0.31	-2.50	-6.34	-1.94	-3.97	2.71	-1.84	-2.30	-0.45	-1.70	-1.02
0.0026	0.84	2.07	2.16	-2.69	-6.71	-5.30	-2.38	-1.06	-0.82	-0.42	-1.08	1.27	-0.82	-0.80	-1.89
0.0028	-3.00	0.38	-1.36	-1.77	0.58	-0.89	0.23	-0.32	-0.80	-0.15	-2.01	-1.20	6.17	4.64	1.33
0.003	0.24	0.45	0.26	-0.64	0.32	-1.14	-3.42	1.27	-1.00	3.62	2.70	1.37	-4.70	-6.87	-6.93

Table 16: Relative Performance of the GA “Majority” trading rule, Transaction costs

The table presents values of Giancomini-White test statistics for comparing the relative out-of-sample performance of the “Majority” rule based on three information sets against the screen information. 99 independent runs of the genetic algorithm have been performed to select the best in-sample trading rules. The “Majority” rule produces a “buy” (“sell”) signal if the majority of the 99 best in-sample rules produce a “buy” (“sell”) signal. This combined rule is then used to trade out-of-sample. Transaction costs are reflected in the bid-ask spread as trading is based on the best bid and ask limit orders. k is the threshold value for the trading band and measured in basis points. Columns $Book - Screen$ contains statistics for relative performance of the limit order book vs. the screen information, $Order - Screen$ compares the order flow vs. the screen information and $Full - Screen$ corresponds to the combined information set vs. the screen information. Highlighted cells contain significant values at 5% level; dark grey colour corresponds to the cases where the $Screen$ information outperforms the corresponding information set, light grey colour presents the opposite situation.

	13-17 January 2003			10-14 February 2003			17-21 March 2003			31 March 2008			1 April 2008		
k	$Book - Screen$	$Order - Screen$	$Full - Screen$	$Book - Screen$	$Order - Screen$	$Full - Screen$	$Book - Screen$	$Order - Screen$	$Full - Screen$	$Book - Screen$	$Order - Screen$	$Full - Screen$	$Book - Screen$	$Order - Screen$	$Full - Screen$
0	3.09	1.78	1.95	0.00	2.16	2.11	0.27	2.74	2.11	1.08	1.18	1.13	0.99	0.99	0.99
0.0002	1.13	0.06	0.19	0.56	0.42	1.12	2.32	0.34	3.91	1.35	1.30	1.65	1.26	10.70	11.34
0.0004	1.60	0.00	0.79	0.51	0.81	0.24	0.68	0.68	1.52	3.54	5.51	0.84	2.00	0.94	0.94
0.0006	5.96	8.52	10.13	2.19	8.76	3.31	1.48	1.40	1.90	1.07	1.87	1.07	0.08	1.55	1.34
0.0008	2.43	3.37	2.03	0.23	0.55	4.44	3.90	0.32	1.93	0.27	2.25	1.77	3.18	3.69	1.72
0.001	1.09	0.33	0.36	1.63	0.13	1.46	1.63	0.03	1.67	0.46	1.08	1.18	3.05	2.23	1.05
0.0012	1.74	0.70	3.24	8.16	0.45	0.44	0.36	1.23	3.36	0.26	0.36	0.01	1.00	5.10	5.74
0.0014	0.78	2.89	2.18	3.94	1.23	7.42	2.73	0.07	1.24	1.16	1.11	4.99	3.69	1.97	2.49
0.0016	5.18	3.47	3.07	3.05	8.52	1.27	0.20	2.94	3.06	N/A	3.70	3.04	N/A	2.00	0.96
0.0018	1.41	0.82	0.52	0.26	6.96	5.06	0.38	0.22	0.22	2.86	0.07	0.85	0.11	2.14	2.14
0.002	0.90	1.38	2.35	6.05	1.80	7.26	0.70	2.58	2.95	0.06	1.01	1.72	2.00	1.96	1.96
0.0022	2.41	2.86	1.42	3.14	1.61	2.05	1.51	0.59	0.41	2.70	3.67	1.32	2.76	2.25	2.97
0.0024	0.33	1.67	0.01	3.43	1.87	4.09	1.17	2.72	3.93	1.75	0.64	1.78	N/A	N/A	N/A
0.0026	5.05	0.01	0.01	1.18	3.70	3.81	0.59	0.86	2.00	2.44	N/A	N/A	2.00	0.04	2.31
0.0028	1.33	0.52	1.66	1.27	3.37	2.76	1.31	1.31	1.31	N/A	N/A	N/A	2.00	2.00	2.00
0.003	0.69	3.13	4.02	2.74	2.55	1.09	2.00	N/A	1.07	2.00	2.30	2.30	N/A	N/A	N/A